Deposon: An Auditable, Conservation-Guaranteed Scattering Layer over LLM Reasoning Paths — II. One Audit Arc, from the Machine Ledger to the Judgment That Reads It

Qihao Yuan

Evidence-strength convention (before any reading). Every claim in this paper carries one of three strength tags, fixed before the reading it qualifies: **closed (pre-registered)** — the decision rule was frozen as a pure function before any run, mechanically evaluated to closure; **consistency** — consistency evidence without pre-registration; **motivation only** — motivation without evidential force. Readers are asked to hold this scale while reading the abstract.

Abstract

The *deposon* is a condensate: a scattering layer over LLM reasoning paths whose every node binding carries two couplings, and whose three scattering channels — transmitted, reflected, and dissipated — sum to identity by construction. The published first paper established the mechanism and its conservation ledger. This paper reports what happened next, as one continuous arc. The question the arc pursues is the one the first paper's value proposition leaves open: the condensate does not promise to be more accurate — it promises to *keep the books*, and to keep them recomputable, step by step, by any third party. What, then, is such a ledger worth, and what does its trust actually rest on? The arc descends through six faces of that question. On the constructive face, the strongest reading of the ledger — the potential-game reading — survives four construction-face legs while its central universal proposition is killed at claim level, coverage 4/22 graphs. On the robustness face, the ledger's readings fail their immunity, recount, and degeneration re-examinations, with root causes registered not as real falsifications but as fake falsifications and uncomputability — propositions untouched, constructions unsolved. On the live face, the layer's meeting with real LLM players is registered (dual calibers 80.0% / 73.3%, co-reported) while the distortion-bound experiment that would formalize the layer's central question stands blocked, its forward payoff values adopted as design inputs (2026-10-12) and its theoretical bounds still undelivered. On the adjudicative face, the arc turns on itself: verdicts are shown to require root-cause attribution before they mean anything — an instrument self-audit recovers a falsified verdict, the same literal PASS carries three different root-cause tiers, and a teacher-metadata probe triggers a real falsification. On the transfer face, the attempt to teach the governing judgment methods to an AI by critical learning fails across three pre-registered rounds (structural similarity 0.0909; critique coverage 0.6842; blind obedience 0.3333; five of six kill-lines in the final round), root causes shifting from construction-dominated to compound. On the substrate face, the judgment ledger's evidence base — an eight-wave, 180-entry collection governed by the same pre-registration discipline — passes its collection-face criteria on two legs and closes known-defects on a third, while a supplementary fourth line fires at its hit line (twelve readings, none reaching the coverage line; closed today on a local chain, the main verdict untouched), with a shared substrate gate closed but not released. The arc ends where it began: both ledgers, machine and human, keep their books under one discipline, and nothing here is a claim stronger than the literal wording of the data.

Keywords: auditability; conservation identity; physics-inspired formalism; pre-registered kill-lines; negative results; chain-of-thought; critical learning; distillation

1 Introduction

1.1 The gap the condensate was built for

Multi-step LLM reasoning currently has two auditable shapes. *Post-hoc provenance* audits a run after the fact, from artifacts the run leaves behind; it can inspect both sides of a run but requires after-the-fact forensics. *Cryptographic proofs* audit inside the run; they are narrow and expensive. Between the two sits an empty layer that neither covers: **runtime, per-instance invariants** — invariants that hold inside the run, at per-instance granularity, without extra cost, and without presupposing an attack model. The first paper of this program [R0] proposed a construction for exactly this layer: the *deposon*, a condensate scattering layer bound to LLM reasoning paths, auditable and conservation-guaranteed, with its game-theoretic face already tested. That paper's compressed value proposition is the seed of everything reported here: the condensate is **not more accurate, but it keeps the books — and the books can be recomputed step by step by any third party with double-precision arithmetic**.

1.2 One arc, not two tracks

This paper is the continuation of the program, and its repository self-declares the trajectory it continues: two completed phases, already published together as the first paper (the mind-map phase and the game-theory-explanation phase), and an ongoing research span whose record this paper is. That span covers two bodies of work — the adversarial validation of the machine ledger, and the critical-learning transfer of the judgment methods that govern the audit — but this paper reports them as **one arc**, because that is what they turn out to be. The machine ledger's validation ends by putting the *judge* on the table: what a reading means is a judgment, and the judgment has its own ledger, collected wave by wave under the same pre-registration discipline, audited by the same recompute rule. The arc of this paper is the full length of that chain: from the books, to the readings, to the verdicts, to the judgments, to the evidence base beneath the judgments, and back to the books.

The arc has six faces, and the faces are the paper's structure. §4.1 asks what survives the strongest reading of the ledger (the constructive face). §4.2 asks whether the ledger's readings hold up under confound control, recount, and degeneration pressure (the robustness face). §4.3 asks what happens when the layer meets real LLM players, and formalizes the central distortion question that remains blocked (the live face). §4.4 asks whether the verdicts themselves can be trusted — an instrument audits itself, a literal PASS is shown to carry three different causes, and a teacher-metadata probe triggers a real falsification (the adjudicative face). §4.5 asks whether the judgment methods that govern this whole discipline can be taught to an AI by critical learning — three pre-registered rounds say no, within their construction domains (the transfer face). §4.6 asks what the judgment's own evidence base is worth, and closes the arc where it opened: the judgment ledger and the machine ledger share one substrate and one discipline (the substrate face).

1.3 What this paper claims, and how it claims it

The claims are organized as a tree with one root. The root is the condensate's value proposition (§2). Under it, six face-claims (A1–A6, one per face of §4) each carry their own verdict literals; the tree is stated in full in §5, and every node is checkable against a sealed on-repo artifact by path, sha256[:12], and byte count (Appendix A). The discipline that makes the claims checkable — pre-registered kill-lines, three-state verdicts, root-cause classification, bidirectional extrapolation bans — is the paper's *instrument*, not its thesis (§3). Negative results are reported as part of the

evidence, at the same weight as closed conclusions; the register of outcomes (§5) is not an apology but a map.

1.4 Contributions

- **A1 (closed, pre-registered).** The potential-game reading of the condensate's ledger closes as a composite: four construction-face legs PASS (co-reported, zero substitution), the central universal proposition FAILs at claim level with coverage 4/22 graphs and a sign-consistent violation set (§4.1).
- **A2 (closed, pre-registered).** The ledger's reading-protection faces close predominantly FAIL/KD with root causes registered as fake falsification and uncomputability — propositions untouched, constructions unsolved (§4.2).
- **A3 (registered).** The live face is measured (dual caliber co-reported) and its central formalization — the distortion bound of the scattering layer over LLM players — is specified, pre-registered, and blocked on the undelivered theoretical bounds, with the forward payoff values adopted as design inputs (2026-10-12; §4.3).
- **A4 (closed, pre-registered; evidence face).** The verdict chain itself is auditable: an instrument self-audit recovers a falsified verdict without touching sealed files; a reporting rule — *no reading enters the text without its root-cause attribution* — is established from a same-literal triad and a teacher-metadata real falsification (§4.4).
- **A5 (closed, pre-registered).** Critical learning of the governing judgment methods fails across three rounds within its construction domains; the root-cause composition shifts from construction-dominated to compound, and a six-model cross-examination places the failure in the construction layer with cross-robust evidence (§4.5).
- **A6 (mixed verdicts, co-reported).** The judgment ledger's substrate passes two collection-face criteria, closes known-defects on a third, and fires a supplementary fourth line at its hit line (twelve readings, none reaching the coverage line; closed today on a local chain, the main verdict untouched); the shared substrate gate closes without release; all divergences are co-reported, none adjudicated (§4.6).

1.5 Paper map

§2 gives the condensate. §3 gives the instrument. §4 is the arc, face by face; each case is reported in four beats — the question, the verdict, the root cause, the boundary. §5 is the audit map: the six faces resolved into what survives, what was killed, and what settled. §6 positions the work. §7 concludes. Appendices A–F carry the evidence ledger, the governance record, the divergence register, the open-gap register, the module fingerprints, and the terminology map.

2 The Condensate: One Ontology, One Ledger

This section is the trunk the whole arc grows from. It is stated once, compactly, in six layers, following the first paper's own order.

2.1 Representation: concept-decomposition graphs

An LLM backend decomposes a task into a directed concept graph G=(V,E) whose nodes are of five kinds — numeric, operation, trap/decoy, answer-candidate, and general-concept nodes. Candidate paths are generated by BFS and ranked shortest-first. The corpus face is published and frozen: 24 files under `corpus/v20/` (16 structural-family graphs, 6 real-semantic-family graphs, index and README), with the index carrying its own fingerprint (Appendix E). The 22-graph judgment corpus used in §4.1 is a subset of this face; four of its graphs are material to the arc's first face.

2.2 Mechanism: the Deposon state and three-channel scattering

Every node binds two couplings — a path coupling g_couple and an aether coupling g_aether (resonance energy E_0) — and every reasoning path is scattered through three channels:

- **T (transmission)** — the path's retained share;
- **R (reflection)** — the path's rejected / rolled-back share;
- **A (dissipation)** — the path's irreversible deposit into the aether: non-volatile bookkeeping with **no recovery channel**.

The scattering formula holds for arbitrary parameters: $\Lambda = 1 + g_eff + g_aether$, with $T = 1/\Lambda$, $R = g_eff/\Lambda$, $A = g_aether/\Lambda$, and $g_eff = g_couple/(1+\delta^2)$ (a Lorentzian detuning factor). Two analytic limit states bracket the construction: the blocking state ($g_aether \to 0$) and the tunneling state ($g_aether \gg 0$), continuously interpolated by $\eta = g_aether/g_couple$. One honesty lock from the first paper travels with the mechanism and is restated here verbatim in force: the scattering form is *inspired by* Feshbach resonance; the T/R/A weights are **constructive definitions, not S-matrix derivations**. The physics vocabulary is a fixed borrow, not a physical derivation.

2.3 Conservation: the ledger

The core invariant is $\mathbf{T + R + A \equiv 1}$, constructive for arbitrary parameters. The per-path energy ledger closes by one-step induction: $E^{(n)} + E_refl + E_diss = E^{(0)}$. The first paper's flagship reading, quoted as published: **maximum per-path energy-audit deviation 2.2e-16 (machine precision)** — rejected paths leave a machine-recomputable record instead of disappearing. Irreversibility is implemented at source level — `dissipate()` accumulates monotonically into a persistent counter, and there is no `recover()` — and argued theoretically: dissipation into an infinite-dimensional orthogonal state space defeats Poincaré recurrence. The first paper's own honest boundary travels with the claim: any finite-dimensional implementation has a Poincaré recurrence time (numerically many orders beyond the age of the universe); theoretical irreversibility and implementational finitude are co-reported.

2.4 Dynamics: reverse dynamics and the potential-game face

On generation tasks (concept-graph completion; leave-one-out prediction; named Hits@3), the reverse dynamics is natural-gradient descent with linear annealing under row-simplex constraints (β over 50 steps, lr 0.1), over a field energy $E(s,t) = -\log(\Sigma_p\ \Pi_e\ t_e) + \lambda_smooth \cdot \Sigma\ W^2{}_{ij}$. The potential-game modeling quartet: each LOO-predicted edge is a player; strategies are candidate target nodes; utilities are field scores; and $\Phi = -E$ is the potential-function candidate. This is the reading whose fate §4.1 reports.

2.5 The published readings the arc inherits (one paragraph)

The first paper is not re-argued here; its published readings are the inherited baseline, quoted as published and re-anchored in Appendix A: the synthetic trap benchmark (unified 100%/100% vs decoy-capture baseline 7%/10%, closed/pre-registered); the label-permutation ablation (17.2% $\pm$ 6.4%, uniform-parameter degradation 10%); the equal-weight decoy control E9.4 (GSM8K 0.85 vs 0.04; StrategyQA 0.899 vs 0.202 — the advantage traced to BFS shortest-path ranking plus trap labels); the rule-filter tie E9.5 (GSM8K 0.87 $\geq$ 0.85, p = 0.5; StrategyQA 0.899 = 0.899, p = 1.0 — "the differential value is in machine verifiability only"); the real benchmarks (GSM8K CoT 97.0 / unified 85.0 / blocking 86.0, McNemar p = 4.9e-4; StrategyQA unified 89.9 vs CoT 92.9, p = 0.549, three arms tied at 89.9); the fusion-dilution negative result (physics 0.484 $\rightarrow$ 0.452, historical 0.783 $\rightarrow$ 0.739 at $\lambda = 0.5$; the λ=2 pseudo-gain registered as a counter-field artifact — any fusion gain must be non-linear); the monotonicity leg GT-5b (22/22, 100% $\geq$ the 80% line, kill-line untriggered, closed); the near-gradient leg GT-6 (median residual 1.594e-29 $<$ 0.10, with above-line disclosure: S4 = 0.148, L_algorithm_process = 0.136, S5 = 0.121); the empirical coordination ratio GT-4 (median 1.333 $>$ 1.2 over 17 finite-valued graphs, 3 infinities, three below-unity readings 0.5/0.75/0.5 co-reported); the formal three-proposition falsification GT_FORMAL (61 graphs $\times$ 338 tasks $\times$ 20 steps = 6,760 kill states; P1a max$\|$T$-$BR$\|\infty$ = 0.8569 with 1$-$cos $\approx$ 0.91; P1b min cos = -1.0, min $\Delta\Phi = -1.2424$e-2 over 10/338 violations; P1c 81-point τ grid all failed, τ^* median 0, 85.98% of states at τ^*=0); the potential-game demotion (cyclic-graph median residual 0.669 $\Rightarrow$ the claim demoted to *approximate*); the domain-partition leg GT-7 (field and prior each win their own domain; hub concentration dominant, OLS $\beta = 2.12$, p = 0.0003, n = 20 exploratory); and the semantic-ablation legs GT-8b/8c (no semantic transfer in rankings; the field is constructive). The team resolution of 2026-10-12 (G-E09) fixes the supply caliber for these inherited readings: no new supplementary-experiment round is opened for the first paper's faces; any reading the body needs is supplied by on-disk recomputation under the three-artifact citation discipline (pure computation, recomputed on demand, 0 pre-run).

2.6 The module substrate

The ontology is implemented as a module family whose contracts are pinned by docstrings and test assertions; all fingerprints are re-anchored in Appendix E. The scattering implementation carries its own self-description at source level — "an AGI-system imitation of physics / a unified-field-theory algorithmic implementation of the condensate" — and pins the node-type semantic parameter table (trap 5.0/0.0; operation 0.3/0.2; others 0.05/0.05). The diffusion prototype implements masked edge completion with a row-simplex projection the source itself labels "the unitary hard projection of T+R+A=1, transposed onto the generation field". The evaluation protocol is zero-I/O, zero-network, shared across the frozen run-scripts. The game-theory base holds the single shared caliber ($\Phi = -$scatter_energy(aggregate)) with per-script frozen copies and a test lock — divergence turns tests red instead of propagating silently. A photonics mapping exists at component-level first-order numerical modeling and is not claimed further here.

The root claim of the paper (T0). *Not more accurate — but it keeps the books, and the books can be recomputed step by step by any third party.* Everything that follows is the audit of what this claim is worth.

3 The Instrument: Pre-Registered Kill-Lines as Governance

The instrument of the arc is a governance layer, stated here once and applied everywhere. Its positioning is a priori and narrow: **the kill-line discipline is a setting of the instrument, not a result.** Three sentences fix its epistemic status. First, a pre-registered criterion specifies the *falsification conditions* of a claim — it does not specify the claim's content; the content comes from the construction and the readings. Second, every claim of this paper can be examined independently of the criteria; the criteria only decide which claims go out. Third, an a-priori setting is not by itself evidence — which is why every closed verdict discloses, side by side, its criterion literal, its freeze timestamp, and its recompute entry.

The instrument's twelve rules, compressed to their operational content: (1) three-state verdicts only — PASS, FAIL, or KD (known defects), no graded language; (2) freeze before execution, effective-locked, no post-freeze amendment, `new_thresholds_added = 0` verified per verdict; (3) verdicts are pure functions of pre-registered criteria; (4) readings are transcribed at face value, connectives banned from verdict statements; (5) every FAIL carries a root cause in a three-way classification — real falsification, fake falsification (the experiment failed, not the proposition), or mixed; (6) bidirectional extrapolation bans at every FAIL and every construction-face PASS — a fake falsification does not kill the proposition, a construction-face PASS does not license a real-face claim; (7) registered divergences are co-reported, never resolved in text; (8) hung items stay hung ("not found → pending PI" is the accepted form); (9) nothing in the paper takes effect by itself — adoption is the PI's action; (10) counting discipline — counts quoted with time points, no stale counts as current values; (11) the verdict vocabulary is time-layered — this paper's judgment faces are all of an earlier date and use the original three-state vocabulary, while a later five-terminal vocabulary is registered as protocol evolution only; (12) negative results are part of the evidence, at the same weight as closed conclusions.

The instrument connects to the first paper's published three-tier evidence scale (closed / consistency / motivation only) by mapping: closed (pre-registered) ↔ the three-state verdict face; consistency ↔ consistency evidence; motivation only ↔ motivation. The narrative follows the published tier vocabulary; the three-state machinery is governance infrastructure, stated in this section and in Appendix B, and is not itself a claim of this paper.

4 The Audit Arc

One arc, six faces. Each case is reported in four beats: the question (which face of the condensate it puts at stake), the verdict (sealed literal, transcribed), the root cause (three-way classification), the boundary (both extrapolation bans). Cases are cited by descriptive name in the text; the sealed artifacts behind every reading are in Appendix A, and the name-to-identifier map is Appendix F.

4.1 The constructive face: what survives the strongest reading

The question. The inherited baseline includes a formal falsification (GT_FORMAL's three equivalence propositions, all killed) and a demotion (the potential-game claim demoted to *approximate* on cyclic graphs). The strongest surviving reading of the ledger is the potential-game quartet of §2.4. The question of this face is exact: under adversarial, pre-registered examination, what part of that reading is alive?

The verdict. The five-case game-theory verdict family closes as **1 FAIL at claim level (kill direction) + 5 construction-face PASS legs + 0 KD**. The FAIL is the family's central claim verdict: its original literal is `inconclusive_preregistered_undefined_band`, sealed as **FAIL**

(claim level, kill direction; A-leg PASS / B-leg FAIL). The five legs — the near-gradient leg, the low-arm re-run leg, the monotonicity leg, the coordination-ratio leg, and the independent-play leg — all PASS at construction face with the sealed-literal rule in force; the co-reporting is total, with zero substitution between legs. Key readings of the claim verdict, transcribed: material-face coverage $4/22 = 18.2\%$ (the construction face contains 4 graphs — S6, L_physics_concepts, L_biological_taxonomy, L_algorithm_process — out of a 22-graph corpus; the other 18 graphs were not judged, and the pre-registration's §9 bans extrapolation); the B-leg literal is a universal proposition ("all graphs"); the four gap values are mutually distinct with the sign criterion 4/4 self-consistent; and of three violations, two have tiny magnitudes — the L_algorithm_process gap of -0.00077 is $1/397.7$ of the S6 gap of -0.30659, the load sitting almost entirely in the S6 graph. The coordination-ratio leg's reading: median $1.3333333333333333 > 1.2$, $n = 17$. The independent-play leg's reading: $0.1 < 0.4 - 0.2 = 0.2$ (strictly holds) and $20 \geq 15$, both conditions met, no undefined band, sealed literal `supports_potential_game_framework` — with the co-reporting note that this leg and the coordination-ratio leg are two verdicts, reported side by side, zero substitution.

The root cause. The claim verdict's root cause is registered as **mixed**, in two layers, transcribed. At the construction-face layer, the B-leg failure is a real falsification — the root-cause triad's class ❶: the reading and tool faces show no failure (the Φ definition, tolerance, and endpoint machinery all recompute), so the violation is in the material face. At the proposition-face layer, the elements for a full real-falsification call (❸❹) are not satisfied — coverage 18.2% against a universal proposition, with two of three violations at negligible magnitude — hence the composite: **class ❶ at the construction face, ❸❹ unmet at the proposition face ⇒ ❸ mixed.** The distinction between the triad's class ❶ and the four-element real-falsification test is registered, not conflated.

The boundary. The claim-level FAIL licenses no statement that the potential-game reading is dead: 18 of 22 graphs were never judged, and the pre-registration itself bans the extrapolation. Symmetrically, the five construction-face PASS legs license no real-face claim: they are construction-face closures, and Rule 6 cuts both ways. What survives, exactly, is a four-legged construction-face reading with a killed universal proposition at 18.2% coverage — the arc's first settled face.

Bridge. A reading that survives at construction face must next survive being *attacked* — by confounds, by recounts, and by degeneration. The robustness face is where the ledger's readings themselves are put under load.

4.2 The robustness face: immunity, recount, and the protection of readings

The question. The ledger's readings — the numbers that make the books auditable — can fail in two ways: the machinery that produces them can be confounded, or the machinery that protects them can degrade. This face holds four case families that attack the readings from those two directions.

The immunity re-examination (two rounds). The question: is the field immune to the confound where decoy count and trap strength are collinear — i.e., does the claimed effect survive decoupling the two? **Round 1 verdict: composite FAIL (α-leg FAIL with IMMUNE 2 / NOT_IMMUNE 10; β-leg FAIL with PASS 15 / BELOW_TOL 3).** The kill-lines were frozen before the experiment by approximately 14 hours (the freeze-to-run interval, with its derived note: 13 h 54 min 59 s). A knife-edge case is disclosed: the α-leg reading 0.04999999999999993 sits in two tiers' band simultaneously — under both the literal and the mathematical reading of the band it falls in IMMUNE, and removing or retaining the knife-edge cell leaves the leg's final state unchanged. **Round 2 verdict: claim-level FAIL (α over 24 cells: IMMUNE 8 / NOT_IMMUNE 16; β over 36 cells: PASS 28 / BELOW_TOL 8, paired registra-**

tion). In round 2 the original collinearity confound is resolved — 36/36 cells satisfy n_decoys == D — while the rank-object mechanism component is fully retained (all 2,880 trap positions read $-\infty$). **Root causes: fake falsification in both rounds** (round 1: the confound-removal component, plus four registered companions — the identical-cell component, where the three-trap tier's field mean is identically 1.000 with zero variance over 6/6 cells × 4 domains = 24/24; the resolution-floor component, where the one-trap tier's threshold of 2 items is twice the granularity of 0.025 so the β-leg's one-trap FAIL sits at the measurement floor; the coverage component (four domains); and the minus-infinity mechanism component, where trap labels absent from the ranking object's node set set the field mean to $-\infty$, so "not deceived" is a mechanism result of the ranking object's domain, not a semantic-recognition claim; round 2: the sampling-implementation component is the common primary cause — the same item under different draws engages different decoys, so the α spread measures decoy-implementation drift, not an immunity breach — with the minus-infinity mechanism fully retained as the common mechanism premise (the original confound resolved, 36/36 cells satisfying the decoy-count equation), accompanied by the resolution-floor and coverage components). A registered divergence is co-reported: the main-reading and a pure-literal reading disagree on the round-1 root cause (fake versus real falsification), resolved by the registered discretion that mechanism-recomputable evidence takes priority — the discretion is registered in full so that any verifier can overturn it independently; an alternative sensitivity reading that removes six identical cells leaves both legs FAIL. A conditional blank was registered for round 2's effectivity (if the batch-9 register were ruled not an effective registration, the verdict would re-rule to KD); the team resolution of 2026-10-12 (G-J01) now rules the batch-9 effectivity **established** — two independent on-disk record points carry the effective state, the batch-9 register's internal contradiction stays on file unrevised, the conditional is not triggered, and the round-2 FAIL final state stands unchanged. **Boundary: both rounds' FAILs license no statement that the field is confounded in fact — the propositions are untouched; the constructions are unsolved.**

The conservation recount. The question: when the ledger's conservation readings are recounted against the sealed originals, do they hold? **The verdict: file-level FAIL, tier 3 (the original annotation stands).** The recount's readings, transcribed: three ledgers give distinct-count readings 9 / 8 / 9; the min/max pairs are O1 0.0132/0.0173, O3 0.3389999999999997/0.3557, O5 17.9501/72.0337. **Root cause: fake falsification (criterion face / caliber face).** A genuine divergence is co-reported: the main reading (criterion-first, mechanism-recomputability-priority) gives FAIL, while a pure-literal reading gives PASS — the divergence is real and registered, with the pure-literal reading not adopted (a placeholder literal without an effective-freeze trace). Four hung items (Q1–Q4) remain hung: the authorization fact-layer evidence is not on disk in any form, and substituting a transcription for the original would fabricate the evidence chain, while substituting a presumption for authorization would exceed authority. **Boundary: the FAIL licenses no statement that the conservation identity failed — it is a finding about the recount's criteria and calibers, not about the ledger's arithmetic.**

The degeneration re-run. The question: after a coordinate-collapse defect is repaired, does the reading-protection machinery recover its discriminating power? **The verdict: overall FAIL.** The line-level split: one line FAILs; seven lines PASS — eight lines in total. The post-repair readings: code count 9; zero pairs 50; the tightest within-class tier reads 1.33, systematically worse than the uniform random control (20/2); the Hamming mean moves 1.961 → 3.026. **Root cause: fake falsification 2 (construction / material face).** The registered caution that travels with the Hamming move: it is offered as a reading, not as a discrimination-recovery claim — a code-length effect is confounded with it. **A supplementary reading face, closed today under a pre-registration-first discipline (freeze at 17:33:58, effective-locked, `new_thresholds_added`**

= 0): the min/max/std reporting face closes with 231 pairs at mean 3.026, zero pairs 50, code count 9, dual-computation pairs 22, byte-hash pairs 22, and within-class four-tier readings 3.6 / 5.33 / 3.4 / 1.33 — with the standard deviation reported in both calibers (the population caliber as main, the sample caliber as alternative, co-reported), a ten-for-ten cross-check against the existing registered values all matching, and the FAIL verdict's final state maintained with zero reopening. **Boundary: the FAIL licenses no statement that the degeneration protection is dead; the discrimination-recovery question remains open with its confound registered.**

The degradation-family verdicts. The question: do the degradation families that watch the readings — three families covering the reading-protection surfaces — close cleanly? **The verdict: all three families close KD at family level;** at face level, 8 faces = KD 4 + PASS 4; across the 19 sealed lines: PASS 12, FAIL 1, KD 6. The sole FAIL targets **a self-certification declaration — not any numeric reading**: the numeric face recomputes 41/41 sequences identically with zero deviation and zero tolerance relaxation (five 1-ULP differences were each proactively self-registered by the declaring role, so the honesty face shows zero concealment); the input face is zero-covered on 4 of 6 faces, and 11 upstream input fields (8 distinct) have zero hits in the declaring role's file. **Root causes: two families fake falsification; the third has zero real and zero fake — uncomputability, hence KD. Boundary: the families' KD licenses no positive claim about the protection surfaces; the single FAIL licenses no statement beyond the declaration face it names.**

Bridge. Every verdict so far was produced by an instrument reading a ledger. Readings are produced by machinery; machinery runs on runners; runners have fingerprints — and one runner's fingerprint is now registered in two values. The robustness face has quietly raised the question the next face must face head-on: what happens when the layer and its verdicts meet the world as it is?

4.3 The live face: real players, and the blocked central question

The question. The constructive and robustness faces are construction faces. The condensate's reason to exist is to sit over *real* LLM reasoning. Two cases put the layer at the boundary: a live-player measurement, and the formalization of the layer's central live question.

The live-player probe. The question: with retrieval augmentation attached, how does the scattering layer perform over real LLM players? **The verdict: a live-run registration, not a three-state verdict face** — the probe's design places it outside the closed-verdict machinery, and it is reported as what it is. Readings, transcribed: 30 cells (15 GSM8K + 15 StrategyQA); the no-retrieval arm closes under **two calibers, co-reported with zero selection: 80.0% (same-extractor re-judged) versus 73.3% (stored original)** — the dual-caliber divergence is registered, not resolved. Six open items are registered as open: the extractor's own status, the cause of two blank cells, a sign-caliber divergence in one component (now ruled by the team resolution of 2026-10-12, G-D09: the post-normalization runner face is the authoritative caliber, the historical reading keeps its registered literal, and the runner-fingerprint dual value stands — the item is closed with zero re-ruling of the live-run registration), a release-wording item, a status-field item, and a cost item — five items remain open. The runner's fingerprint is dual-registered — the verdict file's in-file registration (one value) versus the current disk measurement (a second value, a +1,302 B drift attributed to an authorized modification (a sign-normalization correction), with the registered value present in five on-disk carriers at today's HEAD under the string-search rule (the register's chain-face list co-stated: five faces, two carrying the post-modification value), hence not a fabrication) — with the drift attribution now sealed in a dedicated registration artifact (closed today, zero adjudication of which side is authoritative). **Boundary: the registration licenses no performance claim in either caliber; the dual reading stands as a dual reading.**

The distortion-bound experiment. The question is the condensate's own central question, now formalized: *when LLM players replace Bayesian players, is the total distortion D(M,T) of the three-channel scattering mechanism inside its theoretical lower and upper bounds?* The experiment is fully specified and pre-registered: **kill-line H0 (distortion multiple $\geq$ 2.0×) and closure line H1 ($\leq$ 1.5×), with five anchors (the coordination-ratio reading, the game-theory verdict family, five frozen runs, the LLM client, and the distortion calculator) — any single anchor drifting forces a full withdrawal of the entire version. The verdict: blocked, in full, on the undelivered theoretical bounds** — the theory input's first installment has arrived and been adopted, and the bounds themselves have not. The standing request for the theory input was on file with its five items dispositioned by the team resolution of 2026-10-12 (G-D03); its external channel has now delivered — the commissioned derivation reply, registered in the repository at the remote HEAD `74c7d76` (anchored in Appendix A) — and the PI's adoption ruling of the same day (21:54) adopts its **forward payoff values** — the per-decision optimum `u*(a_t)` = 1 in all 15 mechanism × family cells and the normalization constant `max_u` = 1 in each of the five task families — **with the applicability conditions carried inseparably** (a value-and-condition adoption): the payoff convention is a binary scoring of accepted versus rejected decisions (one accepted-decision reward, payoff range {0, 1}; the normalizer is a per-decision payoff upper bound); the comparator is a full-information, per-decision oracle — for the complete-evaluation family the accepted-action key may contain information withheld from the player, so the player's restricted-information optimum is a different quantity and is not asserted to equal 1, and for sequential games the comparison is regret against the best action at the actual state, not against the payoff of a globally optimal counterfactual trajectory; the premise is a non-empty evaluator key per decision (an unanswerable instance with no accepted action is outside the proposal, and for a mechanism class that removes actions, changes their outcomes, or lets the key depend on its internal state, that state and feasibility must be included and nonemptiness checked before the values apply); and the game-decision family's criterion key must be specified before its cells apply — a Nash equilibrium is not generally a maximum of the players' raw payoffs, best-response against specified opponents and equilibrium recognition are different tasks, and the binary scoring deliberately discards the magnitude of cardinal strategic losses — adoption would choose that loss meaning rather than fill a harmless numerical scale slot; if the intended measure must retain cardinal game losses, this binary proposal is rejected for those cells rather than relabeled as that measure. **The adopted values are forward-looking design inputs: not historical payoffs recovered from the repository, not calibrated findings about any model or mechanism, and not bounds — the theoretical lower and upper bounds are not established by them.** The numerical minima and tolerances remain `null` with their blockers carried as-is: the minima need the instantiated observation laws, admissible policy classes, and keys per cell (the reply's exact conditional minimum is a derivation for a stated exogenous, fixed-horizon policy class, not a value for any existing cell); the tolerances need a mechanism-specific convergence certificate and its connection to regret (a bounded expected rejected-decision count — $0 \leq D \leq H$ over a fixed horizon, no averaging by the number of decisions — is a domain fact, not the requested tolerance). Alongside the values, three further pieces of the same reply are registered as **adopted external derivation content** under the theory-source caliber (the C4 correction register: the theory-bound source is a codex derivation; the team creates 0 theoretical content of its own): two operational definitions — information-channel capacity, measured in bits per channel use over a specified finite observation channel (it does not assign a measured channel to an LLM), and payoff-relevant identifiability, the conditional common-action criterion (an observation is payoff-relevantly identifiable when one accepted action is common to its entire conditional state support; identifying a zero-regret action need not identify the whole mechanism, state, or a unique label) —

the conditional lower-bound derivation (the exact minimum over the admissible observation-based policy class for the stated exogenous class — deterministic choices attain it and randomization cannot improve — attained and therefore a minimum, not a loose nonnegativity bound), and the uniform-mechanism exact check (conditional error probability $1 - |K_t|/4$, and $3H/4$ over a fixed horizon with exactly one accepted action per decision — a result for that specified uniform policy, not a convergence tolerance and not automatically the information-constrained minimum). The slot's status advances from pending-external-input to **adopted** (PI 2026-10-12 21:54); the empirical instantiation remains un-run — recomputing an actual D(M,T) still needs the instance keys, actual choices, decision counts, and a specified weighting law, none supplied — and the team pre-fills 0 numerical L/U bounds (the adopted values are transcribed external derivation content, not team-created bounds). The theory-source ledger that governs where the bound must come from has its own status: the current canonical source is a codex derivation (a single correction layer, pending the PI's re-review as registered — the 2026-10-12 fidelity check verified this draft's source-caliber statements against the correction register with 0 divergence, and the adoption ruling of the same day proceeded on this caliber; the older anonymous layer is retained as history), and the registered caution is quoted in force: *a changed source is not a delivered theory* — with this round's adoption carrying the same discipline one step down, **an adopted forward value is not a delivered bound** — the calculator and the five anchor JSONs have zero hits on disk, so nine preparation items remain unmet and the "uncomputable" trigger state stands. **Boundary: the blocked state licenses no statement about the distortion bound in either direction; the adopted values license no calibrated-payoff or established-bound claim in either direction; the anchor-withdrawal rule is restated so that any future run inherits it.**

An asset-recovery family (verdict, re-run conditions, anchor recomputations, repair recomputations, two alias-restoration artifacts, and one asset recovery) is registered on this face's periphery and anchored in Appendix A; its readings are procedural, and none of them alters any verdict above.

Bridge. The live face ends with a formalization waiting on its undelivered theoretical bounds (the forward payoff values adopted as design inputs, the bounds themselves not established) — and with two calibers of one reading co-reported. Both facts point at the same door: a reading's meaning is not in the number; it is in the judgment that reads it. The arc now turns on that judgment.

4.4 The adjudicative face: can the verdicts themselves be trusted?

The question. The arc so far has trusted its instrument. This face audits the instrument and the judgment behind it — with three cases that share one theme: a verdict literal is not a fact until its cause is known.

The instrument self-audit. The question: when a verdict turns out to have been produced by a mislabeled tool, what does the discipline do? The registered facts: a falsified-verdict case was identified in which the tool face had mislabeled a hash — a "SHA-256 first 16" label fed to SHA-1 hex output; the mislabeling mechanism is fully registered. The action hierarchy was re-ruled by the PI from "kill" to **"recovered"**. The recovery is procedurally sealed: six items are transcribed into a standing ledger; of 15 fingerprint hits, 10 were appended and 5 were skipped — with the registered note that skipped is not missed (the two skipped families are self-referential prefix files that must not be rewritten); the discipline is append-only (zero deletions, zero rewrites); a four-way name-collision check was run; a read-only backup was taken first; zero network calls, zero external LLM calls, zero API use, and zero credentials in plaintext. **Root cause: fake falsification (tool face).** The case is the discipline's own evidence face: the instrument that kills claims also kills

its own errors, on the record. **Boundary: the recovery licenses no claim about any other verdict; every other verdict stands as sealed.**

The same-literal triad. The question: can one verdict literal mean three different things? The triad's three constructions answer yes. **First construction: PASS, registered as false robustness** — a temperature-flip line closes PASS with zero cells hit, but the verdict's own attribution reversal records it as a construction-failure artifact: 49.3% empty responses and 72.1% empty–empty pairs produce a J-statistic of 0.0 as "both-empty agreement" rather than true judgment agreement, under a reasoning model run at a max_tokens = 100 budget mismatch; the real-falsification elements ❷❸❹ are lost; a companion line's 12 cells all FAIL on sample insufficiency. **Second construction: PASS, registered as true robustness** — same caption pair, J = 1.0 same-caption and J = 0.0 cross-caption, i.e., content disagreement where it should be. **Third construction (the re-run): PASS, registered as true robustness with the root cause eliminated** — pair expansion 6 → 15 per teacher with the minimum-sample line at 10; real-falsification elements ❶❷❸❹ all satisfied; zero empty responses (against the first construction's 49.3%); readings: same-caption J median 0.75–1.0 on the stricter scoring arm, 0.5–0.5556 on the looser arm; cross-caption J = 0.0; pooled J = 0.0 (cross-caption dominated, as in the second construction); two companion lines close FAIL with real-falsification attributions (an endpoint's inherent variance; a literal sample floor), prior verdicts untouched. **The registered lesson, adopted as this paper's reporting rule: no measurement reading enters the text without its root-cause attribution. Boundary: the triad's final PASS licenses no robustness claim beyond its own construction; the first construction's PASS licenses nothing at all — which is the point.**

The teacher-metadata probe. The question: the judgment methods this whole discipline runs on are human judgments; is the human teacher side of the collection identifiable from its metadata — and is the teacher quality sufficient? **The verdict: the whole-case direction is a real falsification.** The line-level split, transcribed: the teacher-quality line (five-teacher mean accuracy ≥ 0.70) reads **mean 0.5617 < 0.70** — one endpoint's 0.7512 is the sole item at or above the line, the other four endpoints all below — and is *not triggered* (weakly maintaining the earlier direction); the metadata-identifiability line (pooled AUC ≥ 0.75) reads **0.5663, with a five-fold cross-validation range 0.5266–0.6318, all five reads below 0.75**, and is **triggered — real falsification**; the below-0.60-endpoints line reads 3/5 and is not triggered; the construction lines (non-degeneration, compliance) read pass = True (10 cells all paired). The reproduction face is registered: a whole-chain rerun (5 teachers × 22 captions × ≥5 calls × bidirectional ⇒ ≥1,100 calls; measured 1,090 calls, 99%+ of the lower bound; one teacher endpoint's 77 calls a registered gap disclosed by the worker and read as not a construction failure), with stratified five-fold cross-validation and a linear discriminator over six metadata features. The prior verdict's literal is quoted verbatim with zero overturn. **Root cause: real falsification, elements ❶❷❸❹ all satisfied. Boundary: the endpoint identities are quoted as recorded; the teacher-side entity list is not expanded in this paper; the single strong item is reported with the verdict's own no-extrapolation note — zero "completely indistinguishable" conclusions are licensed, in either direction.**

Bridge. The judgment face closes with a rule — readings need attributions — and a fact: the judgments themselves are human, collected, and finite. The arc's next face asks the boldest question in the program: can those judgment methods be *taught*?

4.5 The transfer face: teaching the judgment method

The question. The judgment methods that govern this discipline — criterion structure, tradeoff patterns, principle application — are human methods. The registered goal of this face is a transferable judgment-method rule set, internalized by an AI for *critical* use. The boundary is stated once, from the registration, and binds everything: this is **critical learning of a way of thinking — explicitly not profiling, not behavior prediction** ("my way of thinking can be critically learned by an AI . . . it should not be turned into a profile of me" — the registration's authority line, translated from the registered original 「我的思维方式可以被 AI 批判性学习……不应是给我做画像」). The falsifiable claim is threefold: structural similarity ≥ 0.80; critique coverage ≥ 1.00; blind obedience ≤ 0.10. The first round's caliber (reproduction-consistency rate) is deprecated, and the deprecation is disclosed here by rule rather than hidden.

The verdict chain (three rounds, all FAIL). *Round 1 — FAIL (exploratory)*, with the fake-falsification suspicion unexcluded (construction-failure family); three kill-lines fired; the round's UNK rate read 55.6% and its permutation p 0.2937. *Round 2 — FAIL (formal)*: all four kill-lines fire. The readings, transcribed: structural similarity **0.0909** (n = 22 held-out; alternative caliber 0.1618 at n = 17); critique coverage **0.6842** (19 divergent items, 13 critiqued; alternative 0.6923); blind obedience **0.3333** (3 agreement / 1 blind; alternative 0.0); the bootstrap line — lower bound ≥ 0.50 — reads **CI [0.0, 0.2045]** (n = 1000; alternative [0.0294, 0.3382]). Root-cause composition: **2 compound + 2 fake-falsification-class + 0 pure construction-infeasibility.** *Round 3 — FAIL (formal)*: five of six kill-lines fire; the sentinel dual-caliber consistency line does not trigger. The round-3 threshold set was frozen under the same discipline (structural similarity 0.65 in caliber A and 0.55 in caliber B; critique coverage 1.00; blind obedience 0.10; bootstrap lower bound 0.40; plus the sentinel). The readings, transcribed: structural similarity closes under two co-reported measures — the alignment-based measure reads **0.4534** against the 0.65 line and the normalized-edit-distance measure reads **0.2754** against the 0.55 line, both FAIL (alternative readings 0.5127 and 0.2963); critique coverage **0.4286** (alternative 0.3636); blind obedience **0.2222** (9 agreement / 2 blind; alternative 0.2857); the bootstrap line reads **CI [0.3217, 0.5870]** (alternative [0.3722, 0.6611]) against the 0.40 line; permutation p **0.0160**. Root-cause composition: **3 compound + 2 fake-falsification-class + 1 sentinel PASS + 0 pure construction-infeasibility** — the attribution has shifted from construction-dominated to compound across the chain, and the construction displacement is itself registered (mechanical binary → composite attribution → construction displaced, FAIL persisting).

The six-model cross-examination. The question: is the chain's failure a property of one student, or of the construction? **The verdict: across six models, all five lines FAIL in all six — 6/6 × 5/5 — with every attribution in the tool/construction layer** (three literal branches: degenerate pair distance; dual-arm zero difference; proxy proportion < 0.20). A model-upgrade control (one model swapped for its successor) does not overturn: same root cause, FAIL maintained. This is cross-robust evidence that the failure sits in the construction layer.

Boundary. The three-round FAIL chain licenses no statement that the judgment methods are not learnable in general; the kill verdicts are construction-domain-limited, and the cross-examination itself locates the failure in the construction. Symmetrically, nothing here licenses any claim about the teacher beyond the probe's own boundaries (§4.4).

Bridge. A construction-layer failure is still a failure of something built — and what was built was a dataset, collected wave by wave under the same pre-registration discipline that governs the machine ledger. The last face of the arc examines that substrate.

4.6 The substrate face: the ledger beneath the judgment

The question. Every judgment in the transfer face was learned from a collected ledger: human judgment events, wave by wave, pre-registered, on disk. The final face asks whether that ledger's own collection face holds — and finds, at the bottom of the arc, the same discipline that started it.

The collection. The substrate: **180 entries over six days, collected in eight waves** (the day-caliber divergence is co-reported: the event-date field reads 6 days; an any-date-string rescan reads 11 days — two readings, zero replacement). The main wave is a pure-question instrument (80 questions); a supplementary open wave adds 8 items, all of the thought-oriented kind (题性: 思考型 — registered literally). The closing inventory: present 19 / absent 0 / mismatch 3 (three discrepancy identities registered). The privacy discipline is absolute and restated: **reasoning full texts are never quoted in this paper**; dataset JSONs are cited by path, SHA-12, and bytes only.

The collection-face adjudication. The question: does the substrate pass its own pre-registered collection criteria? **The verdict: three lines close — two PASS, one KD; the line closes KD overall.** Transcribed: the cross-day line ($\geq$ 5 days) reads **6 days, PASS** (an alternative domain reading of 11/13 days is co-reported); the single-day-share line ($\leq$ 60%) reads **0.5000–0.5333, PASS** (tightest margin 0.0667; an older 56.5% value is not reused — the verdict's own literal rejects it); the front-half line ($\geq$ 50 entries) reads **160–180, PASS**. The correction/reversal line ($\geq$ 5 events) is the KD: seven registered readings (2 / 5 / 7 / 8 / 11 / 16) straddle the threshold — the is_correction=true count reads 7, the correction_confirmed=true count reads 2, and 35 entries lack the is_correction field outright; the backfill rule that would resolve the field gap is a draft now ruled in by the team resolution of 2026-10-12 (G-D04) as the execution caliber for the held-out segment's future adjudication — the collection verdict's KD final state is unchanged (the rule applies to the segment's new readings, not to this verdict's on-file readings). Root-cause count: real 0 / fake 0 / KD 1 / PASS 3. Two sensitivity analyses are registered, neither adjudicated: the thin-day analysis (2 of 6 days are thin; removing them flips the cross-day line to 4 days $\Rightarrow$ FAIL; the flip threshold is exactly the removal of the 2 thin days), and the reverse-sensitivity analysis with dual denominators (adding the 13 entries of a later date, 96 $\rightarrow$ 109 with N $\rightarrow$ 193, gives 0.5648 — still $\leq$ 0.60; holding N at 180 and reaching 108 gives 0.6000 — exactly on the blade). A fourth criterion line (vocabulary-expansion) was in a locked, unverified state closing **zero-verdict** (ceiling transcripts 0.7143 main / 0.6000 alternative, no-direct-source mark); the team resolution of 2026-10-12 (G-D05) locks the fifteen-word table into the locked set, with two words' tier assignment carried as dual readings (both tiers read, zero selection), unlocking the line's adjudication — and the line's supplementary adjudication has since run to **full chain closure as a local artifact pair** (pre-registration frozen 18:48:09, effective-locked, the thirty-nine-word lock's effective time-point team-designated; verdict registered; both pushed at commit `74c7d76`, add-only). The ruling, transcribed from the verdict file: **the hit line fires — the registered hit-flag reads True (FAIL, the criterion's hit line; the flag's registered identifier maps in F.3)** — with four dual-caliber readings, the base thirty-six-word face reading **0.571429 / 0.500000** and the locked thirty-nine-word face reading **0.714286 / 0.700000**, all below the 1.00 coverage line; twelve registered readings (including the pre-repair arm and the contrast arm) show **none $\geq$ 1.00, so the conclusion is fully robust**; arithmetic self-check 12/12 (including the index face); no re-run of the executor — the registered readings were adopted read-only; the collection verdict's main file was SHA-rechecked identical before and after the run (**zero touching, zero reversal, zero retro-edit**). Three dual-value faces co-report with zero resolution: the measured ceiling face (0.8571 / 0.8000) versus the briefing's transcribed values (0.7143 / 0.6000); the alternative caliber's definitional divergence (the parent file defines 10 words, the registered line carries 11 —

unreproducible); and the briefing's transcribed values themselves. Root cause: the construction face's beta component, now locally repaired in part (two genuinely missing items at indexes 47/50, plus the table's structural ceiling — the 1.00 structural unreachability stays as the on-file literal). The verdict file's own fingerprint history is disclosed: a correction wave moved 35,022 → 35,037 bytes with verdict tier and readings unchanged. **Boundary: the KD licenses no statement that the collection is adequate or inadequate; the sensitivity flips are registered as flips, not adjudicated.**

The shared substrate gate. The question: both ledgers — the machine ledger of §2 and the judgment ledger of this face — draw on one shared substrate; is the gate that protects it from degeneration closed? **The verdict: the gate's three gap-faces are fully closed; the gate line closes KD; closure is not release.** Transcribed: the cross-day line PASS (7 days); the share line PASS at **0.516129**; the front-half line PASS (186 entries; four reading domains, the lowest reading 74); the back-half line **KD — the reading is 1 < 5 and the reading authority is not in place**; the overall line KD; the minimum-sample line PASS at 74. Six readings are co-reported for the gate (the main reading adopted; a three-day domain reading would FAIL and is reported-not-adopted; two readings show zero divergence; one reads KD; the old 56.5% value is rejected by literal). Two substrate-domain divergences are co-reported and unresolved: the adjudication ledger counts 180 entries / 6 days / 0.5000–0.5333 while the gate's recompute reads 186 / 7 days / 0.516129 — two sealed artifacts, different domains, zero reconciliation — now ruled by the team resolution of 2026-10-12 (G-D02): the full-event domain (the seven-day caliber) is the authoritative measurement domain, and the legacy three-day accounting domain is retained as a co-reported contrast reading (its FAIL is a contrast reading, not a re-ruling; both verdicts' final states stand); and 39 entries lack the event_id field (an honest-accounting item, not decidable here). A three-caliber count (77 / 74 / 62) that the gate depends on now has a dedicated consolidation register on disk (closed today: the three calibers are listed side by side, zero selection; the gate's three prerequisites are now all in place per the same resolution set (the three-caliber consolidation register on disk; the fifteen-word table locked by G-D05; the closure-is-not-release caliber confirmed by G-D08), so the gate stays closed-not-released — the release action remains a literal-source-checked act, and nothing in this paper releases anything). **Boundary: the closed gate licenses no run-permission claim; release is the PI's action, and nothing in this paper releases anything.**

The held-out segment. The substrate's stratified held-out segment is registered and deferred: its criterion pre-registration is now frozen and pushed (at commit `74c7d76`; execution scheduled for the next window), the registered three-line inconsistency conflict and the codebook-collapse risk (23 → 50+ cells) remain on file, and zero execution, zero verdict stands.

Bridge. The gate is closed, not released; the collection passes where it can and settles where it must; and the discipline at the bottom of the judgment ledger — pre-registration, freeze, recompute, co-report — is the same discipline that governs the machine ledger at the top. The arc closes where it opened: one condensate, one discipline, two ledgers.

5 The Audit Map

The six faces resolve into a map. Following the condensate's own identity, the map's columns are what the arc's verdicts scattered into: **transmitted (survives), reflected (killed), dissipated (settled as boundary)**. The map is a presentation of sealed literals — no claim is added by the arrangement.

Face	Transmitted (survives, construction face)	Reflected (killed)	Dissipated (settled / blocked / KD)
Constructive	5 legs PASS (near-gradient, low-arm re-run, monotonicity, coordination-ratio, independent-play)	claim-level FAIL, universal proposition, coverage 4/22	—
Robustness	degeneration re-run: 5 lines PASS; recount: numeric faces recompute	immunity rounds 1–2 FAIL; degeneration re-run overall FAIL; degradation families: 1 declaration-face FAIL	degradation families KD ×3; recount divergence (FAIL vs PASS, co-reported); Q1–Q4 hung
Live	live-player registration (30 cells; dual caliber 80.0/73.3)	—	distortion-bound experiment blocked (forward payoff values adopted as design inputs, theoretical bounds undelivered); 6 open items; runner fingerprint dual-registered
Adjudicative	triad construction 3 PASS (true robustness); teacher probe construction lines PASS; self-audit: recovered	teacher probe: metadata line TRIGGERED (real falsification); triad companion lines FAIL	self-audit action hierarchy re-ruled (kill → recovered); same-literal triad: attribution is mandatory
Transfer	sentinel line not triggered	three-round FAIL chain; six-model cross: 6/6 × 5/5 FAIL	construction-layer attribution (cross-robust)
Substrate	collection lines: cross-day, share, front-half PASS; gate lines: 4 PASS	supplementary vocabulary-expansion line fired at its hit line (twelve readings, none $\geq$ 1.00; local chain, main verdict untouched)	correction/reversal line KD; gate overall KD (closed, not released); held-out segment unfrozen

What the FAILs mean. Under the root-cause discipline, a FAIL with a fake-falsification root cause is a finding about the experiment; a FAIL with a real-falsification root cause is a finding about the world; a KD is a finding that the question could not be asked cleanly. The arc contains one real falsification at a proposition face (the constructive face's B-leg, class ❶), one real falsification on the metadata line of the teacher probe, and a chain of fake-falsification and uncomputability findings everywhere else. **What the FAILs do not mean.** The bidirectional bans cut both ways, and this paper reports them as load-bearing: the transfer chain's FAIL is construction-domain-limited and licenses no "not distillable" claim; the teacher probe's real falsification licenses no "completely indistinguishable" claim; the degeneration FAIL licenses no "the protection proposition is dead" claim; and symmetrically, no construction-face PASS — the five legs, the gate's four lines, the collection's three lines, the triad's final construction — is extrapolated to any real-face claim.

The verdict is not the reading. The adjudicative face's triad is the map's emblem: one literal, three causes. The reporting rule it produced — no reading without root-cause attribution — is the discipline's answer to its own hardest problem, and this paper applies it wholesale.

The arc's shape. The arc began with a condensate that keeps books, and ended with the

discovery that keeping books is a two-sided discipline: a machine ledger that cannot be trusted without audits, and a human judgment ledger that cannot be taught without constructions. Between them stands one instrument — pre-registered, frozen, recomputable — and one boundary sentence, quoted (with an ellipsis where the registry block condenses) from the repository's own README and left as the map's caption: *negative results are reported as part of the evidence ... readings, preregistrations and audit records stay on disk, and nothing here is a claim stronger than the literal wording of the data.*

6 Related Work

Game theory, potential games, and the price of anarchy. The constructive face's machinery stands in the potential-game lineage [R1–R8]: ordinal potential games and congestion-game foundations; the price-of-anarchy program [R4–R8] frames the coordination-ratio leg (median 1.333 > 1.2, published) and its below-unity co-readings. The formal falsification of the three equivalence propositions (published) is the continuation's inheritance. Algorithmic game theory and mechanism design [R9–R12] supply the adversarial-analysis frame this paper's instrumentation inherits; the empirical-coordination-ratio construct is registered as distribution-level and explicitly not the classical ratio.

LLM reasoning, its fragility, and its measurement. The live and transfer faces join the measurement-of-reasoning literature [R20–R29]: chain-of-thought prompting and its stepwise variants; fragility findings on GSM-style symbolic perturbation [R24] and simple-task breakdown [R25]; faithfulness measurement [R26–R27]. This paper's contribution to that conversation is methodological and negative: the same-literal triad (§4.4) is a concrete demonstration that a robustness literal can be an artifact of construction, and the reporting rule it yields (attribution before reading) is offered to that literature as practice.

Negative results and reporting norms. The paper's genre stands with pre-registration and reporting-reform work [R30–R37]: the trouble with benchmarks and dataset drift; distribution shift; the reproducibility program. The closest external relative is pre-registration for predictive modeling [R53], which shares the falsification-conditions-first stance; the present instrument differs in committing to three-state verdicts with mandatory root-cause classification and in reporting the governance layer's own failure (§4.4's self-audit) as part of the evidence.

Distillation and chain-of-thought learning. The transfer face's contrast anchors are knowledge distillation [R42–R43], step-by-step distillation [R44], reasoning-teacher approaches [R45], and instruction-conditioned synthetic data [R46]; the on-policy line [R47] — students learning from self-generated errors — and the imperfect-teacher line are the closest external framings of the critical-learning goal. The sycophancy benchmark [R48] is the external reference frame for the blind-obedience construct, with the registered direction distinction attached: that line measures model-toward-user pandering; this paper measures student-toward-teacher blind obedience — different direction, same mechanism family. The sycophancy-measurement critique [R49] (existing measurements conflate related behaviors and lack operational definitions) is external support for the paper's stance in §3: adherence metrics require operational definitions frozen before measurement. One related-work slot on behavioral profiling and its limits remains open by scan, and is stated as open.

The first paper's lineage (semantic priors and knowledge-graph completion). The published baseline's knowledge-graph-completion face [R38–R41] is inherited as published context and is not extended here.

7 Conclusion

This paper reported one audit arc over one condensate. The arc's headline inventory: a potential-game reading alive at construction face with its universal proposition killed at 18.2% coverage; reading-protection faces failing with causes registered in the experiment, not the propositions; a live face registered in two calibers while its central formalization waits on its undelivered theoretical bounds (its forward payoff values adopted as design inputs, its bounds not established); a verdict chain audited by its own instrument and ruled by attribution; a transfer program that failed three times in its construction domain with cross-model evidence placing the failure in the construction; and a judgment ledger whose substrate holds under one shared discipline with a gate closed but not released. What survives the arc is the discipline itself: pre-registration before readings, three states without gradation, root causes before meanings, divergences without adjudication, and both extrapolation bans at every verdict. The condensate keeps the books. This paper is part of the evidence that keeping them is worth the audit.

(Read-before-cite rule applied: entries carrying a "pending verification / as registered" note are scan-level faces and are not load-bearing for any claim; the 12-entry verification-pending list from the scan remains pending and is not cited beyond face level. The FTFB artifact is registered by fingerprint only (Appendix A), 0 content description.)

Appendix A — Evidence ledger (path + sha256[:12] + bytes)

Conventions. Paths beginning `results/`, `docs/`, `reviews/`, `README_V4` are relative to `_september_workspace/` (the mirror-clone location of the registered remote paths); paths beginning `paper/` and `letters/` are relative to the repository root; paths beginning `deposon_team/` are relative to the workspace root. Values marked ✔ were independently re-measured at the archived drafts' measurement pass (2026-10-12, clone HEAD `f44996fbfd9f386299442924f70585bb20d935fc`; 65 anchors re-measured, 63 matched; the 65-anchor pass counted 2 divergences — E20 and the archived table's backfill-rule anchor — while this ledger's dual-value rows number seven in total (E20, E26, E28, E58, E59, E72, E73): three carry in-ledger divergence faces (E20, E26, E28) and the other four are registered-epoch contrasts (E58, E59, E72, E73), all co-reported with 0 adjudication; the review-side measurements were taken at later clone HEADs (`72c5839...{}`, and today's `c247c157eced...{}`), and all drift-class anchors are dual-registered so that a reader re-measuring at any later HEAD can compare against both values). Values marked [m] were re-measured for this v2 repair round at clone HEAD `c247c157eced...{}` (this seat's own `sha256sum`/`stat` pass; 0 fabricated values, 0 carry-over of unparseable paths — the v1 draft's placeholder-style path family was wholesale replaced by the on-disk true paths in this round, basis: the archived 65-anchor table, the closing reviews' measured values, and this pass). Values without a mark are registered measurements transcribed with 0 re-measurement. **This rewrite's own full-anchor re-measurement pass is scheduled as the next round (per the commissioning instruction); the anchor rows added today: five newly closed (E67–E71 — all pushed by the 21:1x window: remote main `f44996f...{}` → `9de6fd7...{}` → `c247c15...{}` → `74c7d76...{}`, every commit add-only, verified by `git ls-remote`; E71's pair pushed at `74c7d76` and re-measured in place at the remote HEAD this round) plus two gate-distribution faces (E72/E73, added per the re-verification with dual values — a git-traceable historical blob and the current disk value — both re-measured in place) plus one letters/ adoption anchor (E74, added this round: the codex unblock reply, the adopted forward values' source artifact — pushed at `5e865e3`,**

in force at remote HEAD `74c7d76`, re-measured in place this round, the same-batch trio co-verified). The remote-registered column carries git blob sha1 values only where they are the sole registration form. Case-identifier column: the descriptive names of the body text map here to the registered identifiers (full map in Appendix F).

#	Face / case (descriptive name)	Artifact	Anchor
E01	constructive: the five-case verdict family	`results/_v5_gt_verdict_2026_09_29.md` = `45648b0ec940` / 57,227 B	✔ [m] GT five-case verdict
E02	constructive: family pre-registration (claim-level criterion literal)	`results/_v5_gt5_gt6_gt8_gt5b_gt_prereg_2026_09_28.md` = `2505ff5d203f` / 63,447 B	✔ [m] GT pre-registration
E03	constructive: historical-JSON re-attribution carriers (byte-untouched)	`results/deposon_v20_gt5.json` = `fca14c5735dd` / 72,647 B; `_gt6.json` = `04b90b638bdc` / 65,105 B; `_gt8.json` = `2b88948dca19` / 5,396 B; `_gt5b.json` = `2907006dbe48` / 56,425 B; `_gt.json` = `497b9c2d6746` / 3,803 B	✔ GT historical JSONs
E04	robustness: immunity re-examination, round 1 verdict	`results/_v5_item35_v2_verdict_2026_09_29.md` = `b59bef293712` / 26,232 B	✔ [m] (M1/R1/R2 readings registered in-file)
E05	robustness: immunity re-examination, round 2 verdict	`results/_v5_item35_v3_collinear_verdict_2026_10_01.md` = `363cce3dae59` / 38,266 B	✔ [m]
E06	adjudicative: instrument self-audit, six-item ledger	`results/_fake_verdict_reverify_sixitem_ledger_2026_09_30.md` = `41d7b76ea4d6` / 40,957 B	✔ [m]
E07	adjudicative: instrument self-audit, session records ×2	`results/_v5_fake_verdict_session_log_2026_09_28.md` = `132bfa0c02f5` / 18,877 B; `results/_fake_verdict_session_log_reverify_2026_09_30.md` = `c14094bdf815` / 25,953 B	✔
E08	adjudicative: instrument self-audit, pre-audit registration tier	`results/_v5_fake_verdict_v1x_prereg_2026_09_28.md` = `088085d6c90d` / 28,484 B	✔
E09	robustness: conservation recount, verdict	`results/_v3_recheck_26b_verdict_2026_10_01.md` = `8bbecb15b28b` / 26,309 B	✔ [m]

#	Face / case (descriptive name)	Artifact	Anchor
E10	robustness: conservation recount, verdict-prep	`results/_v3_recheck_26b_verdict_prep_2026_10_01.md` = `3aab3f7660fc` / 36,133 B	✔
E11	substrate: shared substrate gate closure and recompute verdict	`results/_t1_gate_closure_and_recompute_verdict_2026_10_01.md` = `caf20d1fa538` / 72,926 B	✔ [m]
E12	live: distortion-bound V0 specification	`docs/V3X/P_B_DISTORTION_BOUND_V0_SPEC.md` = clone `aab9d78ada2d` / 15,549 B (local original 15,457; historical 11,189 / 7,126 — drift registered, C-12)	✔
E13	robustness: degradation-family verdicts	`results/_v5_pjpm_pc_verdict_bang3_2026_09_29.md` = `1814f9590f0f` / 41,566 B	✔ [m]
E14	transfer: registered goal statement + falsifiable-claim family + frozen threshold tables	`results/_v4_pi_cot_v2_prereg.md` = `cc25c5149ce1` / 4,247 B (the v2.2 goal literal's authoritative carrier; threshold-table set)	✔ [m]
E15	transfer: round-1 verdict (exploratory)	`results/_v4_pi_cot_v2_verdict.md` = `eb9ad4193cf2` / 22,340 B	✔ [m]
E16	transfer: round-2 verdict (formal)	`results/_v4_pi_cot_v2_verdict_v2.md` = `5d79e67a4e9d` / 33,723 B	✔ [m]
E17	transfer: round-3 verdict (formal)	`results/_v4_pi_cot_v3_verdict_v3.md` = `bb44fdc7ab0f` / 53,751 B	✔ [m]
E18	transfer: round-3 pre-registration (threshold set, sentinel line)	`results/_v4_pi_cot_v3_prereg.md` = `b7547329af2e` / 27,223 B	[m]
E19	live: with-RAG live-run data	`results/_v3_n_cpath_live_data_2026_09_27.json` = `25374af92f88` / 12,711 B	✔ [m]
E20	live: runner-copy fingerprint (dual value)	`deposon_team/plugins/_v3n_cpath_live_rerun_2026_10_01.py` = in-file registration (10-01) `3e932a5409b5` / 20,831 B vs current disk `7d8d26174ce9` / 22,133 B	✔ [m] dual-registered (C-26)
E21	live: live-rerun verdict (release record)	`results/_v3n_cpath_live_rerun_verdict_2026_10_01.md` = `a91efbfec3d0` / 16,409 B	✔ [m]
E22	adjudicative: same-literal triad, first construction verdict	`results/_v4_supp_t1_verdict.md` = `0ad42625f333` / 29,106 B	✔ [m]

#	Face / case (descriptive name)	Artifact	Anchor
E23	adjudicative: same-literal triad, second construction verdict	`results/_v4_supp_t15_verdict.md` = `3cb5c2f59961` / 54,377 B	✔ [m]
E24	adjudicative: same-literal triad, third construction verdict	`results/_v4_supp_t15r2_verdict.md` = `963725dbd82a` / 65,638 B	✔ [m]
E25	transfer: six-model cross-examination verdict	`results/_v4_track2_multimodel_verdict_2026_09_23.md` = `44d9bad9039b` / 12,883 B	✔ [m]
E26	substrate: collection-face adjudication	`results/_s3_kv3s_3_verdict_2026_10_01.md` = 10-01 `b3ea36a24dcf` / 35,022 B vs current `e394bb6b5ddf` / 35,037 B	✔ dual-registered (10-12 correction wave over-wrote; verdict tier and readings unchanged)
E27	substrate: whole-ledger recompute	`results/_s3_kv3s_3_full_recompute_2026_10_01.md` = `b0123f161c10` / 22,183 B	✔ [m]
E28	substrate: missing-field backfill-rule draft	`results/_s3_missing_field_backfill_rule_draft_2026_10_01.md` = registered `2e77a13e51f1` (per the archived table) vs current `8496a53d183a` / 35,827 B	✔ [m] dual-registered
E29	adjudicative: teacher-metadata probe verdict	`results/_v4_supp_l14v3_n26_verdict.md` = `df6453d57f8a` / 65,978 B (older registered 64,485 — drift disclosed)	✔ [m]
E30	adjudicative: teacher-metadata data face (caption-surface carriers)	`results/deposon_p_d_b3_merkle_22_caption_2026_09_16.json` = `b5873afb9d29` / 18,104 B; `_v0_4_2026_09_29.json` = `33ad1c01f745` / 20,398 B; `_v0_4_2026_10_09.json` = `95f3f0957be8` / 21,176 B	[m]
E31	adjudicative: methodology template (the judgment method's write-up face)	`results/_v4_supp_t15r2_methodology_template.md` = `6f9b865280a3` / 17,411 B	[m]
E32	robustness: degeneration re-run, group verdict	`results/c5_b_group_verdict_2026_10_01.md` = `3fbf6b38a383` / 30,118 B	✔ [m]
E33	constructive: the published first paper (authority copy)	`paper/deposon_paper_final_en.md` = `2ec20b96e941` / 68,164 B	✔
E34	constructive: first-paper v1 copy (registered in this ledger)	`paper/deposon_paper_v1.md` = `18f600782478` / 85,624 B	✔ [m]

#	Face / case (descriptive name)	Artifact	Anchor
E35	constructive: repository README (trajectory anchor, transcribed in App F)	`README.md` = `a80b3db006a5` / 7,279 B	✔
E36	constructive: first-paper continuation family	`paper/v2/deposon_paper_v2X.md` = `77b44d365342` / 59,174 B; `paper/v2/outline_v2X.md` = `116d193b1607` / 10,476 B	✔ [m]
E37	constructive: GT executor file	`results/_v5_gt_exec_2026_09_29.md` = `9a679f43f796` / 27,701 B	✔
E38	robustness: immunity round-1 pre-registration	`results/_v5_item35_prereg_v2_2026_09_28.md` = `1fb2419c6e96` / 35,633 B	✔
E39	robustness: immunity round-2 line-setting file	`results/_v5_item35_v3_collinear_prereg_2026_09_29.md` = `873652fd39dd` / 42,504 B	✔
E40	robustness: immunity round-2 independent recompute	`results/_v5_item35_v3_collinear_recompute_2026_10_01.json` = `a210e2c41275` / 49,682 B	✔
E41	robustness: immunity round-2 cited decisions register	`results/_v5_confirm_b9_decisions_register_2026_09_29.md` = `8c510cda6014` / 17,008 B	✔
E42	robustness: degeneration pre-registration	`results/_v5_v3_deg_fix_prereg_2026_09_29.md` = `98286cc1aec7` / 41,622 B	✔
E43	robustness: degradation-family pre-registration	`results/_v5_pjpm_pc_deg_prereg_2026_09_29.md` = `989ee2bce660` / 59,446 B	✔
E44	live: prior live verdict (byte-untouched declaration)	`results/_v3_n_recheck_llm_verdict_2026_09_27.md` = `426cfe18cf65` / 21,332 B	✔
E45	live: simulation report	`docs/V3X/CPATH_SIMULATION_REPORT_2026_09_10.md` = `de772cd9e7ba` / 6,825 B	✔
E46	substrate: protocol threshold-literal unique source	`results/_v3_s_prereg_v1_2026_09_27.md` = `ef5a40554960` / 46,836 B	✔
E47	live: theory-source caliber correction	`results/_c4_theory_source_caliber_correction_2026_09_30.md` = clone `1f058d9cb7a8` / 25,138 B	✔

#	Face / case (descriptive name)	Artifact	Anchor
E48	live: theory-source intake-acceptance register	`results/_c4_reply_intake_acceptance_register_2026_10_01.md` = remote blob `21ad9581c0ec...{}` / 33,728 B (the sole registration form is the remote blob)	registered
E49	transfer: round-2 signoff	`results/_v4_pi_cot_v2_verdict_v2_signoff_2026_09_26.md` = `ba4d07bd7000` / 26,637 B	✔
E50	transfer: round-3 signoff	`results/_v4_pi_cot_v3_verdict_v3_signoff_2026_09_27.md` = `41d29f4deb70` / 42,157 B	✔
E51	substrate: collection brief (ceiling readings)	`results/_v3_s3_collection_brief_2026_09_28.md` = `d5ba1161e37f` / 19,045 B	✔ [m]
E52	substrate: questionnaire v1 (six-rule protocol)	`results/_v4_pi_cot_v2_questionnaire_v1.md` = `7b14cdb31d21` / 12,402 B	✔
E53	substrate: closeout recompute (closing inventory's counts registered in-file)	`results/_d8_closeout_protocol_recompute_2026_09_30.md` = `57aba2cd26a6` / 31,007 B	✔
E54	substrate: dataset family (privacy face, fingerprint-only citation)	`results/_v4_pi_cot_v3_dataset.json` = `5118f5b44f17` / 16,025 B; `_v4_pi_cot_v2_dataset.json` = `7b01cd835a41` / 12,672 B; `_v4_pi_cot_v3_dataset_addendum_d8_sq_2026_09_30.json` = `769ef8b71462` / 374,072 B; `_v4_pi_cot_v3_dataset_addendum_s3_d1fill_2026_10_01.json` = `b9026cbf5ca2` / 46,909 B	✔
E55	substrate: collection ledger (original-wording preservation)	`results/_d8_collection_ledger_2026_09_30.md` = `031dd8429048` / 20,290 B	✔
E56	live: theory-input acquisition request	`results/_v5_theory_input_acquisition_request_2026_09_29.md` = `19099ecb3c46` / 10,751 B	✔
E57	live: theory-source register family	= E47 + E48 (the caliber-correction file and the intake-acceptance register; no third file exists in the registry)	cross-referenced

#	Face / case (descriptive name)	Artifact	Anchor
E58	substrate: main-wave questionnaire (dual value)	`results/_v4_pi_cot_v3_questionnaire_v2_d8_2026_09_28.md` = 10-01 `238077d25919` / 53,581 B vs current/remote `7656be1d287f` / 53,583 B	✔ [m] dual-registered (anonymization + correction waves; question-face readings unchanged)
E59	substrate: supplementary open wave	`results/_d8_supplement_open_questions_2026_09_30.md` = 10-01 `579b375e9508` / 24,833 B vs current/remote `aa07872e4cab` / 24,835 B	✔ [m] dual-registered (as E58)
E60	substrate: closing inventory	the closing inventory's counts (present 19 / absent 0 / mismatch 3) are registered in E53's file; the bucket register `results/_v4_closeout_bucket_register_2026_10_01.md` = `99355c24a03b` / 41,943 B carries the bucket face	✔ [m] cross-referenced
E61	transfer: round-2 chain (dataset / coding review / signoff)	dataset = E54; coding review `results/_v4_pi_cot_v2_coding_review_2026_09_26.md` = `5fbec21e0ad2` / 42,914 B; signoff = E49	✔ [m] cross-referenced
E62	transfer: round-3 chain (dataset / activation)	dataset = E54 (addendum faces); activation `results/_v4_pi_cot_v2_prereg_activation_2026_09_23.md` = `00584e5a5c78` / 1,016 B	✔ [m] cross-referenced
E63	genre: FTFB prior artifact	`results/_archive_2026_09_20/fiction_that_feeds_back.pdf` = `2d9edc8c7303` / 356,123 B	✔ fingerprint only, 23 pages — existence registration, 0 content citation (read-before-cite rule); registered divergence between the commission materials ("remote not directly accessible") and the clone mirror's presence of the file — co-reported, 0 adjudication registered
E64	genre: arXiv:2609.09001 PDF (existence registration)	metadata face; PDF body 0 read (the on-disk authority copy E33 is the read baseline; the two versions' identity is unverified, registered)	

#	Face / case (descriptive name)	Artifact	Anchor
E65	references: verification and scan records	`docs/REF_VERIFICATION_v2.md` = `64f0a3772c12` / 5,860 B; `reviews/literature_scan_v2X_A.md` = `d0bba9379b4b` / 18,876 B; `reviews/literature_scan_v2X_B.md` = `313b30815a39` / 30,468 B	✔
E66	references: transfer-track README (release record)	`README_V4_2026_09_29.md` = `15648ca22a00` / 58,646 B	✔
E67	**robustness: degeneration re-run supplementary reading face (closed today)**	preregistration `results/_c5_kv3df_0_1_minmaxstd_suppl_prereg_2026_10_12.md` = `88cb33b53a7e` / 5,697 B (freeze 2026-10-12 17:33:58, effective-locked) + readings file `results/_c5_kv3df_0_1_minmaxstd_readings_2026_10_12.json` = `04f14cd2fdbf` / 7,193 B (distribution: min 0 / max 8 / population-std ≈ 2.6300 / sample-std ≈ 2.6357, co-reported calibers) + report file `results/_c5_kv3df_0_1_minmaxstd_suppl_report_2026_10_12.md` = `31a4e2976dab` / 8,476 B (three-artifact chain, prereg-first, `hashlib.sha256` full chain, 0 modification; mtime-reset by owner normalization registered in the report's §4)	registered (pushed)
E68	**live: runner fingerprint dual-value register (closed today)**	`results/_e20_runner_sha_drift_dual_value_register_2026_10_12.md` = `3cee8e81a66e` / 4,352 B (0 selection; drift attribution sealed)	registered (pushed)
E69	**substrate: three-caliber consolidation register (closed today)**	`results/_q2_77_74_62_three_caliber_consolidation_register_2026_10_12.md` = `cc75f1e3d8af` / 4,609 B (77 / 74 / 62 side by side, 0 selection)	registered (pushed)

#	Face / case (descriptive name)	Artifact	Anchor
E70	**constructive: line-setting-file drift closure register (closed today)**	`results/_gt_reconstruction_sha_drift_dual_value_register_2026_10_12.md` = `261ebba5c09e` / 4,789 B (the theory-request adjacent line-setting file's drift, dual value 0 selection; C-13's post-closure caliber; pushed in the fourth commit `c247c15`)	registered (pushed)
E72	**substrate: gate-distribution face 1 (the ruled authoritative face, resolution G-D01)**	`results/_krj10_reverdict_after_r4_convergence_2026_10_11.md` = the historical blob at commit `df51dc3` (the 10-12 upload batch carrying the 10-11 artifact) `a21915ec55a7` / 31,672 B vs current HEAD `3b00b6377dab` / 31,652 B [m] (drift via the `b31d070` correction wave: freeze-then-overwrite + abs-path correction + desensitized re-push, git-traceable) — count-level readings (closed 13/13: pass 7 / kd 6 / hung 0) identical across both faces — the per-item distributions remain the C-7 swapped pair — and carried by the current file, unaffected by the drift	dual-registered, git-traceable
E73	**substrate: gate-distribution face 2 (the coexisting historical face, resolution G-D01)**	`results/_lim_aperiod_adoption_and_label_correction_2026_10_11.md` = registered (10-11) `f4b129e3312f` / 14,770 B vs current HEAD `22e48303c8ad` / 14,750 B [m] (same correction-wave epoch) — count-level readings identical to face 1 (13/13: pass 7 / kd 6 / hung 0; the per-item distributions remain the C-7 swapped pair)	dual-registered, git-traceable

#	Face / case (descriptive name)	Artifact	Anchor
E71	**substrate: vocabulary-expansion supplementary adjudication (closed today; pushed at commit `74c7d76`)**	pre-registration `results/_s3_kv3s_3_4_judgment_prereg_2026_10_12.md` = `a80d9ae8008b` / 5,678 B [m] (frozen 18:48:09, effective-locked; the 39-word lock's effective time-point team-designated) + verdict `results/_s3_kv3s_3_4_supplementary_verdict_2026_10_12.md` = `338f30ad7460` / 6,156 B [m] — **pushed at commit `74c7d76` (the 21:1x window, add-only); fingerprints per the executing seat's registration (§9) and re-measured in place at the remote HEAD this round**	registered (pushed at `74c7d76`)
E74	**live: the adopted external derivation reply (forward payoff values; this round's adoption object)**	`letters/_codex_c4_values_unblock_reply_2026_10_10.md` = `a975c7990cd8` / 21,486 B [m] — **pushed at commit `5e865e3` (the four-file letters batch, add-only, byte-faithful), in force at remote HEAD `74c7d76`; fingerprint re-measured in place at the remote HEAD this round; the same-batch codex trio co-verified in place (`8e4972352291` / 13,378 B, `97a35f8ae94c` / 4,490 B); the adoption object per the PI's ruling of 2026-10-12 21:54**	registered (adopted; PI 2026-10-12 21:54)

Appendix B — Governance record

B.1 Verdict vocabulary time layering

All judgment faces reported in this paper are earlier-dated faces, quoted in the original three-state vocabulary (PASS / FAIL / KD). The five-terminal vocabulary introduced later by the PI (`pass / af / kd / fail / give-up`; overlap priority `af > kd > give-up`; non-five-terminal outcomes count as hung items, not verdicts) governs later-dated faces and is reported here only as protocol evolution. No later-dated verdict face is covered in this draft.

B.2 Terminology compliance (the applied register)

1. The first paper is described only as an "arXiv preprint"; 0 occurrences of "published / accepted / merged / peer-review passed" for it. 2. Target vehicle: arXiv preprint, cs.AI primary classification; 0 journal template, 0 designated submission target, 0 claim of submission intent. 3. The prior preprint's abstract and negative-results face: existence registration only. 4. Unit/affiliation literals: pending PI final review; 0 transcription; author block is a placeholder. 5. **Descriptive-name discipline (this rewrite's own rule, per commission):** the body text (§1–§7 and the Abstract) contains 0 occurrences of lineage labels, main-line numbering, and registered identifiers (threshold-table IDs, judgment-line IDs, wave labels, task names, case numbers); all such identifiers live in the appendices and map to the body's descriptive names via Appendix F. Scan result: **0 hits** (the mechanical scan of §1–§7 for the banned identifier families; the only permitted literals are the first paper's published experiment names, listed in Appendix F's whitelist). 6. The four core principles are registered as preceding and running through the first paper's publication practice; 0 statement that they originate from its chapters. 7. The FTFB artifact is registered by fingerprint only; 0 content description; its side files are neither cited nor used as templates. 8. Dataset/privacy faces: reasoning full texts 0 quoted; dataset JSONs cited by path + SHA-12 + bytes only. 9. All existing files were read-only for this draft; 0 modification, 0 renaming, 0 relocation (this rewrite is a newly created file; the archived drafts v0.1–v0.3 are untouched). 10. Counts are quoted with time points; historical counts are not used as current values. 11. Desensitized entity literals: 0 restoration; mapping literals not reproduced; the desensitization-relevant entity families are not named. 12. 0 fabrication of remote sha1 / path / file counts / clauses; unexecuted actions are marked "not executed"; 0 claims of upload/push/remote-write/receipt beyond the transcribed registration artifacts (the push record of today's three artifacts is transcribed from the executing seat's registration, not asserted as this draft's own action). 13. Third-party conclusions quoted in quotation form; 0 use of any drafting-collaboration agent identity as an attribution name; agent identities do not appear in the paper. 14. 0 delivery-channel instructions; no archive-zone material; 0 credential handling. 15. Verdict vocabulary per the time layering of B.1. 16. Conclusions are checkable (path + SHA-12 + bytes); negative conclusions carry root causes; divergences are co-reported; gaps are stated; this draft's own fingerprint is not written into the draft. 17. Internal sensitive faces (including the teacher face) are not transcribed anywhere in this draft.

B.3 Hard-constraint compliance (H1–H12)

H1 (0 secrets/credentials/session ids/agent rosters/personal information) — compliant; H2 (attribution by quotation; 0 impersonation) — compliant; H3 (0 fabrication; read-before-cite; "not found → pending PI" accepted) — compliant (FTFB and the arXiv PDF are the two unread faces, marked as such); H4 (0 structural constraints imposed externally; the single-arc structure is this draft's own design per the approved outline) — compliant; H5 (kill verdicts unsoftened; banned connectives not attached to any verdict statement) — compliant; H6 (bidirectional extrapolation bans at every FAIL and every construction-face PASS) — compliant; H7 (0 threshold adjustment; all threshold literals quoted from the pre-registrations) — compliant; H8 (privacy faces: path + SHA-12 + bytes only) — compliant; H9 (0 automatic effectivity; adoption is the PI's action) — compliant; H10 (counting discipline with time points) — compliant; H11 (0 half-conclusions; all registered divergences co-reported) — compliant; H12 (0 touching of existing files; this draft is a newly created file) — compliant.

B.4 The 17 factual corrections carried into this rewrite (register)

1. Live-run data path: `results/_v3_n_cpath_live_data_2026_09_27.json` (the one-character path error is not repeated). 2. Runner fingerprint dual-value registration + divergence row (E20/E67). 3. Teacher-quality line = five-teacher **mean** ≥ 0.70 (not "$\geq$ 1 endpoint"); reading mean $0.5617 < 0.70$, one endpoint 0.7512 sole ≥ 0.70, four below. 4. KD used univocally as *known defects*. 5. The verifiable-learning survey entry carries the ledger's "(Proof-of-Learning)" literal. 6. Theory-source canonical = "a codex derivation" named (older layer retained anonymous). 7. Immunity round-1 dual readings include the alternative sensitivity reading (both legs still FAIL after removing six identical cells). 8. The two mechanical root-cause clauses are carried in full (identical-cell $24/24 = 1.000$ zero variance; resolution-floor threshold = twice the 0.025 granularity). 9. The claim verdict's construction-face layer (triad class ❶) is distinguished from the four-element proposition-face test. 10. Freeze interval quoted as "approximately 14 hours" with the derived note (13 h 54 min 59 s). 11. The registration's authority line carries the translation marker from the registered original 「我的思维方式可以被 AI 批判性学习……不应是给我做画像」. 12. The supplementary wave's question kind = thought-oriented (题性：思考型), registered literally. 13. The reverse-sensitivity analysis carries dual denominators ($96 \to 109$ with $N \to 193 \Rightarrow 0.5648$; N held at 180 reaching $108 \Rightarrow 0.6000$). 14. The receipt-tier acknowledgment state remains: not acknowledged. 15. Anchor re-measurement claim caliber = 65 measured / 63 matched / 2 divergent (this rewrite's own pass is the next round; today's additions: five newly closed anchor rows transcribed from registration (all five pushed by the 21:1x window — the pair at `74c7d76`, re-measured in place at the remote HEAD) plus the two gate-distribution faces (E72/E73) entered with dual values re-measured in place; this v2 round additionally re-measured the repaired path family in place, marked [m] in Appendix A; the v8 round adds one letters/ adoption anchor (E74, the adopted external derivation reply, re-measured in place at remote HEAD `74c7d76` this round)). 16. HEAD dual-timepoint registration (the archived pass at `f44996f...{}`; today's pushes at `9de6fd7...{}` and `c247c15...{}`; 0 mixing). 17. Related-work slot-status sentences are synchronized with the actual slot state (one open slot stated as open).

Appendix C — Divergence register (all entries unresolved, 0 adjudication)

#	Face	Reading A	Reading B (. . .)	Status
C-1	substrate gate cross-day domain	whole-event domain: 7 days ⇒ PASS	old accounting domain: 3 days ⇒ FAIL	**ruled by resolution G-D02**: the whole-event domain (7 days ⇒ PASS) is the authoritative measurement domain; the old accounting domain (3 days ⇒ FAIL) is retained as a co-reported contrast reading (FAIL as contrast, 0 re-ruling; both sealed verdicts' final states stand) — the verdict file's pre-registered no-softened-phrasing discipline remains in force
C-2	collection adjudication vs gate recompute	gate: n_after = 186 / 7 days / 0.516129	collection: 180 entries / 6 days / 0.5000–0.5333	two sealed artifacts, different domains, co-reported — **ruled by resolution G-D02: the full-event domain (7-day caliber) is authoritative; the legacy 3-day domain is a co-reported contrast reading (FAIL as contrast, 0 re-ruling)**
C-3	immunity round-2 effectivity	batch-9 register §1.1 "effective"	batch-9 register L11/L197 "pending review"	internal self-contradiction, 0 reconciliation; **ruled by resolution G-J01: effectivity established (two independent on-disk record points); the register's internal contradiction stays on file unrevised; the conditional blank is not triggered — the round-2 FAIL stands**
C-4	conservation recount file-level verdict	main reading FAIL	pure-literal reading PASS	genuine divergence, co-reported; pure-literal not adopted (placeholder literal, no effective-freeze trace)
C-5	three-caliber count	77	74 / 62	0 selection; today's consolidation register (E68) lists all three side by side, 0 adjudication

#	Face	Reading A	Reading B (. . .)	Status
C-6	a matrix-attribution item	—	—	**ruled by resolution G-D10 (scope-of-application interpretation, append-only registration): the nominal three-valued classification axis falls outside the numeric-degradation alert's per-field scope (n_distinct = 3 is full-class coverage, not collapse); the three affected family verdicts' final states are unchanged, 0 criterion-text amendment**
C-7	a gate prerequisite's authoritative distribution	distribution A	distribution B (swapped)	**ruled by resolution G-D01: the authoritative distribution is the `a21915ec55a7` face (closed 13/13: pass 7 / kd 6 / af · fail · give-up 0 each, hung 0); the mutually exclusive distribution of the `f4b129e3312f` §5 face is retained as a coexisting historical time-point record (0 revision, 0 invalidation declared; as corrected by the resolution erratum, the ruling's face designation is unchanged and its per-item value chain is carried by the parent pair's final values) — both faces anchored as E72/E73 with dual values (git-traceable historical blob + current disk)**
C-8	legacy paper-file fingerprint	`8baa9854b3f7` / 239,700 B	third-party self-report `fd39cf811232` / 236,425 B / 1,396 lines	attribution 0 asserted (line-ending/ encoding/version/ carrier all possible)
C-9	two protocol-carrier string items	reading 1	reading 2	not verified, pending PI

#	Face	Reading A	Reading B (. . .)	Status
C-10	degeneration re-run unclosed items	protocol-version carrier gap	min/max/std unreported	**the latter face closed today (E66); the carrier gap remains registered (→ PI; → worker + verifier)**
C-11	39 entries without event_id	"186 = 186 distinct entries"	—	not decidable (the gate verdict's honest-accounting item 9)
C-12	distortion-bound spec byte size	local original 15,457	11,189 / 7,126 historical; clone 15,549	drift registered, 0 adjudication
C-13	a reconstruction doc's fingerprint	line-setting registration `570b7bd3e286` / 6,695 B	on-disk `cff15d000d47` / 13,424 B	drift by another role, registered; **closed today by a dedicated closure register** (`261ebba5c09e` / 4,789 B, commit `c247c15`, E70): dual value retained, 0 selection, 0 reopening
C-14	main-wave instrument fingerprint	10-01 registered `238077d25919` / 53,581 B	current / remote `7656be1d287f` / 53,583 B	anonymization + correction waves; question-face readings unchanged
C-15	supplementary wave fingerprint	10-01 registered `579b375e9508` / 24,833 B	current / remote `aa07872e4cab` / 24,835 B	as C-14
C-16	collection-adjudication fingerprint	10-01 registered `b3ea36a24dcf` / 35,022 B	current `e394bb6b5ddf` / 35,037 B	correction wave; verdict tier and readings unchanged
C-17	teacher-probe verdict size	older registration 64,485 B	current 65,978 B	difference registered
C-18	backfill-rule draft fingerprint	brief-registered `2e77a13e51f1`	this rewrite's transcription source `8496a53d183a` / 35,827 B	divergence registered, 0 adjudication; draft pending PI in any case
C-19	FTFB remote accessibility	commission materials: "not directly accessible; attachment required"	clone mirror: file present	0 adjudication; content unread either way
C-20	literature-scan entry count	self-declared 40	counted 41	registered, 0 adjudication
C-21	ledger date caliber	event-date field: 6 days	any-date-string rescan: 11 days	two readings co-registered, 0 replacement
C-22	correction-event count	is_correction=true: 7	correction_confirmed=true: 2; pair-closure ≥ 5 (fields None)	expanded by the collection verdict to seven readings ⇒ KD; 2↔7 undecided at the collection layer
C-23	live-player no-retrieval dual caliber	same-extractor rejudged 80.0%	stored original 73.3%	co-reported, 0 selection
C-24	live-player sign caliber	report §2	runner literal (one sign convention apart)	**ruled by resolution G-D09**: the post-normalization runner face is the authoritative caliber; the historical reading keeps its registered literal (E20 dual value maintained, 0 selection)

#	Face	Reading A	Reading B (. . .)	Status
C-25	two upload-registration waves	461-file inventory @ `f44996fb...{}`	830-file ledger @ `fb0ee996...{}`	two waves, 0 mixed use; full mapping not established (267-entry intersection registered)
C-26	live-player runner-copy fingerprint	in-file registration (10-01): `3e932a5409b5` / 20,831 B	current disk (measured 2026-10-12): `7d8d26174ce9` / 22,133 B	drift +1,302 B attributed to an authorized modification (a sign-normalization correction); registered value appears in five on-disk carriers at today's HEAD under the string-search rule (the runner, the verdict file, the normalization file, the closure register, and the wrap-up note — the counting artifact itself excluded; the register's own chain-face list names five faces, two of which carry the post-modification value — both calibers co-stated, not a fabrication); dual-value registration (E67/E68), 0 adjudication

Appendix D — Open-gap register (four classes; positions updated to the single-arc structure)

Class A — paper-claim faces (close this week): A-1 distortion-bound theory input (→ §4.3; the five request items are dispositioned by resolution G-D03, and the state is advanced by the PI's adoption ruling of 2026-10-12 21:54: the commissioned external derivation has delivered — the reply in the repository's letters/, in force at remote HEAD `74c7d76` (Appendix A, the E74 row) — and its forward payoff values are adopted with the applicability conditions carried inseparably (`u*` = 1 in 15 cells, `max_u` = 1 in the five families, the numerical minima and tolerances `null`), so the slot advances from pending-external-input to **adopted** (PI 2026-10-12 21:54) while the empirical instantiation remains un-run (instance keys, actual choices, decision counts, and a weighting law still unsupplied), the landing stays assigned to the direct-execution line, now at the pre-registration stage (v9 micro-sync: the delivered mathematics' acceptance verification — independent re-derivation and finite enumeration — passed on 2026-10-12, registered locally pending repo anchoring, so the landing's live gate is the pre-registration freeze), and the adversarial face stays queued on genuine inputs; the team creates 0 theoretical content and pre-fills 0 numerical L/U bounds — the adopted values are transcribed external derivation content, not team-created bounds); A-2 theory-source content delivery (→ §4.3; per G-D03 and the same ruling: the commissioned derivation's item-by-item verifiable delivery has arrived in letters/ — the originated values with in-letter proofs, the two operational definitions, the conditional lower-bound derivation, the uniform-mechanism exact check, the finite-enumeration verification note — and is

adopted as external derivation content under the theory-source caliber; the calculator and five-anchor pre-registration still await their landing, now as components of the pre-registration freeze (v9 micro-sync per the A-1 note: the acceptance verification passed on 2026-10-12, the landing's live gate is the freeze of the pre-registration requirements)); A-3 the claim verdict's material-face coverage extension (→ §4.1; new measurement face requires a fresh pre-registration, 0 amendment of the sealed one); A-4 the low-arm leg's independent measurement face (→ §4.1; same discipline); A-5 the held-out stratification segment (→ §4.6; criterion pre-registration frozen and pushed at `74c7d76`, execution in the next window; codebook-collapse risk on file).

Class B — governance faces (post-week): B-1 the gate's three remaining prerequisites (→ §4.6; **all three now in place per the resolution set — E68 consolidation, G-D05 lock, G-D08 closure-confirmation — release stays a literal-source act, 0 release claimed here**); B-2 the vocabulary-expansion line's verification (→ §4.6; **ruled by G-D05 and now closed by today's supplementary adjudication (local chain, hit line fired, twelve readings all below the coverage line, main verdict untouched — E71, pushed at `74c7d76`)**); B-3 the missing-field backfill rule (→ §4.6; **ruled in by G-D04 as the segment's execution caliber; this verdict's KD unchanged**); B-4 the immunity round-2 effectivity ruling (→ §4.2; **ruled by G-J01: established; conditional not triggered**); B-5 the recount's four hung items (→ §4.2; **resolution G-J02: held on file — the authorization fact-layer is unverifiable on disk, quoted with an unverifiable mark, 0 fabricated authorization**); B-6 a gate-prerequisite distribution assignment (→ Appendix C C-7; **ruled by G-D01**); B-7 the live face's six open items (→ §4.3; **the sign-convention item closed by G-D09, five remain open**); B-8 the transfer chain's follow-up measurement faces (→ §4.5).

Class C — verification faces: C-1 the FTFB and arXiv-PDF unread faces (existence registration only); C-2 the two literature-scan slots (one open slot stated as open; the 12-entry verification-pending list stays pending); C-3 the first paper's v2X continuation material (registration only, 0 body read).

Class D — decision faces (reserved to PI; per the PI's directive of 2026-10-12 — "resolve the pending-PI items yourselves; much historical residue is no longer recoverable" — entries with no on-disk evidence are closed as unresolvable historical residue rather than re-submitted, except D-4/D-5 which belong to the submission workflow and stay open): D-1 the substrate-gate domain divergences (→ §4.6 / Appendix C C-1, C-2, C-11); D-2 the three-caliber selection (0 selection, register E68); D-3 the legacy fingerprint attribution (Appendix C C-8); D-4 author/affiliation literals (→ title block, pending PI); D-5 the paper's final vehicle and submission logistics.

Appendix E — Module fingerprints (the ontology's implementation substrate)

Module	Path	SHA-12	Bytes
Scattering implementation	`_september_workspace/deposon_agents.py` (directory prefix shared by all rows below; sha256[:12] + byte count re-measured for this v2 round)	`5496954b0ad5`	104,622

Module	Path	SHA-12	Bytes
v1.3 shim	_september_workspace/deposon_agents_v1_3.py	715026b48678	1,827
Diffusion prototype	_september_workspace/deposon_diffusion.py	fe4d668eb1da	19,171
Evaluation protocol	_september_workspace/deposon_protocol.py	ca756668ce2e	2,845
G2 modes	_september_workspace/deposon_g2_modes.py	6a175a264b46	6,316
Game-theory base	_september_workspace/gt_common.py	425b4b2f1c01	3,601
Photonics mapping	_september_workspace/deposon_photonics.py	c7e78ad9f030	10,402
Corpus construction	_september_workspace/mindmap_corpus_v20.py	e77e7f3455e0	25,791
Corpus index	_september_workspace/corpus/v20/index.json	8423ffe266af	7,335

Note (registered hash coincidence): the corpus index and one agent-side file share the same SHA-12 prefix `8423ffe266af` at the same byte count — two distinct files whose full sha256 values coincide in the first 12 hex characters; citations must carry the full path. Three architecture-level files are registered as on-repo but unread at the body level (a related-work continuation file, a figure-language policy, a paper brief); 0 content from them appears in this draft.

Appendix F — Terminology and identifier map (registry layer)

F.1 Mapping principles. (i) The body text (§1–§7, Abstract) uses descriptive names only; lineage labels, main-line numbers, and registered identifiers appear 0 times in the body. (ii) Registered identifiers are preserved exactly once per row in this registry layer (Appendices A–F), where descriptive names map to them. (iii) Names already published in the first paper are whitelisted for body use (F.4). (iv) The repository's README trajectory anchor is dual-tracked: the body's descriptive paraphrase (§1.2) plus the condensed frame transcription below (F.5); the on-disk README remains the verbatim authority, reachable by the fingerprint in F.5.

F.2 Face and case map (descriptive name → registered identifier).

Body descriptive name	Registered identifier / old label
the condensate / the machine ledger	the Deposon scattering layer (module self-description: condensate unified-field-theory implementation)
the adversarial validation of the machine ledger (the validation span)	the game-theory main line (V3)
the critical-learning transfer of the judgment methods (the transfer span)	the distillation main line (V4)
the potential-game verdict family (five cases)	the GT five-case family (gt5 / gt6 / gt8 / gt5b / GT-4 / GT-1 legs)

Body descriptive name	Registered identifier / old label
the claim verdict (constructive face)	gt5 (claim level, `inconclusive_preregistered_undefined_band`)
the immunity re-examination, rounds 1–2	#35 v2 / #35 v3
the conservation recount	#26b
the degeneration re-run	C5 (K-V3DF line set; today's supplement = the min/max/std face, K-V3DF-0-1)
the degradation-family verdicts	P-J / P-M / P-C (K-PJPC-0-1 line)
the live-player probe	#32 (CPATH)
the distortion-bound experiment	P-B (spec V0)
the theory-source ledger	C4 (canonical correction layer)
the external derivation reply / the adopted forward payoff values	the codex unblock reply (letters/, the E74 row; the values-commission slot family: the payoff-optimum and normalization-constant slots adopted, the minima and tolerances null)
the alias-recovery family	#21 family (7 artifacts)
the instrument self-audit (falsified-verdict case)	the fake-verdict audit (six-item ledger)
the same-literal triad	T1 / T1.5 / T1.5r2 (K-T1-S3 line)
the teacher-metadata probe	N-26 (K-N26-1/2/3/N1/N2)
the critical-learning verdict chain (rounds 1–3)	Task B v1 / v2 / v3 (K-V2-a…c, K-V3-A…F line sets)
the six-model cross-examination	Track 2
the collection-face adjudication	S3 (K-V3S-3-1/2/3/4)
the held-out stratification segment	S4 (unfrozen)
the eight collection waves / the main wave / the supplementary open wave	D1–D8 / D8 / the SQ supplement
the shared substrate gate	the T-1 gate (K-V3S-3-1/2/3-4, TH-v2-1)
the frozen threshold tables (rounds 2–3)	TH-v2-1…8 / TH-v3-10…14 (TH-COT-v2/v3)
the registered study family (the judgment-ledger program)	the pi_cot registered family
the independent-play leg (§4.1's co-reported pair)	the GT-1 leg (`supports_potential_game_framework`) — distinct from the published GT-8b/GT-8c names, no collision
the low-arm re-run leg	the lowarm r1 / r2 verdicts (gt8's low-arm re-runs; the near-gradient leg is the published GT-6 face, carried by §2.5, not this row)
the family's suspended-items registry	GT-33 (γ-registry face)
the family's pre-registered re-run request	P-10 (the same-discipline new-measurement-face request)
the degeneration re-open chain	P-1 (the degeneration-family re-open chain, §2.53–2.54 of the clue book)
the request-chain artifacts (robustness face)	RJ5 (the re-run request family)
the conclusion-verification probe	the concl_t verification (supplementary probe face)
the falsification-conditions-first instrument	the pre-registered kill-line protocol (12 rules)

F.3 Verdict-line and threshold-line map (body descriptive phrase → registered line ID). The vocabulary-expansion hit-flag line = `k_v3s_3_4_hit` (E71's verdict file); the structural-similarity line = K-V2-a / K-V3-A; the critique-coverage line = K-V2-b1 / K-V3-B; the blind-obedience line = K-V2-b2 / K-V3-C; the bootstrap line = K-V2-c / K-V3-D; the permutation line = K-V3-E; the sentinel line = K-V3-F; the minimum-sample line = TH-v2-1 / the gate's N_min line; the teacher-quality line = K-N26-1; the metadata-identifiability line = K-N26-2; the cross-day / single-day-share / front-half / correction-reversal lines = K-V3S-3-1 / 3-2 / 3-3 / 3-4; the degeneration lines = K-V3DF-0-1, 0-2, 5-1, 5-2, 5-3, 9-x; the distortion kill/closure lines = H0/H1 (P-B spec). All threshold literals quoted in the body (0.80 / 1.00 / 0.10 / 0.50; 0.65 / 0.55 / 1.00 / 0.10 / 0.40; 0.70 / 0.75 / 0.60; ≥5 days / ≤60% / ≥50 / ≥5; 2.0× / 1.5×; N_min) are transcribed from these registered lines with 0 adjustment (H7).

F.4 Published-name whitelist (permitted in the body by lineage). GT-4, GT-5b, GT-6, GT-7, GT-8b, GT-8c, GT_FORMAL, E9.4, E9.5, GSM8K, StrategyQA, Mc-

Nemar, arXiv:2609.09001, the corpus graph identifiers (S4, S5, S6, L_physics_concepts, L_biological_taxonomy, L_algorithm_process), and the first paper's figure/table names. These are published experiment names of the first paper, not registered identifiers of the present program. **Same letter-form note (per the mechanical-gate review): S4/S5 share their letter-form with other faces' case labels; in the body they appear only inside the published GT-6 readings (the above-line disclosure) and are disambiguated by context — all of the body's own case references use descriptive names, so the whitelist is the sole source of these letter-forms.**

F.5 README trajectory anchor (condensed transcription). The repository README (`README.md` = `a80b3db006a5` / 7,279 B) self-declares four main-line frames, transcribed here in condensed form (frame labels and status marks preserved; intra-frame explanatory prose condensed) as the anchor behind §1.2's paraphrase: "V1 脑图主线 ✓ published; V2 博弈论解释 ✓ published; V3 博弈论主线 active; V4 蒸馏主线 active（方法迁移——明示非行为画像）". The first two frames are published as arXiv:2609.09001 (cs.AI; 23 pages, 5 figures). The README's closing note of its published-findings block, transcribed with the same condensation (and quoted in §5 as the map's caption, where the ellipsis marks the condensation): "Negative results are reported as part of the evidence. . . . readings, preregistrations and audit records stay on disk, and nothing here is a claim stronger than the literal wording of the data." **This registry block is the only place in the paper where the lineage labels appear; the body's §1.2 paraphrase is the body's sole trajectory statement.**

F.6 Round numbering. The body's "round 1 / 2 / 3" (the verdict chain; the immunity re-examination) maps to the registered v1 / v2 / v3 chain-of-verdict numbering; the rounds are verdict-chain rounds, not lineage labels.

Appendix G — Repair-round disposition (v2; item-by-item against the three closing reviews)

Basis. The three closing reviews (the reference-and-terminology closing report, 15 items; the game-theory-face review, 13 items; the distillation-face review, 9 items — 37 items with overlaps). Disposition classes: **[R] repaired in this round / [D] self-decided with record (per the delegation order) / [W] deferred pending team resolution / [N] not adopted, with reason**. Overlapping items are dispositioned once under their earliest review.

Item (review · severity)	Disposition	Where / reason
phantom path family, 27 no-value rows (reference report · high)	[R]	Appendix A rebuilt: 70 anchors, true paths + sha256[:12] + bytes, each re-measured this round at HEAD `c247c15` (marked [m]); basis = archived 65-anchor table + the reviews' measured values + this seat's own pass
path mismatches: collection-verdict / backfill-draft / teacher-probe rows (reference · high)	[R]	corrected to `results/_s3_kv3s_3_verdict_2026_10_01.md` (dual value), `results/_s3_missing_field_backfill_rule_draft_2026_10_01.md` (dual value), `results/_v4_supp_l14v3_n26_verdict.md`
directory-level paths, questionnaire/supplement/closing rows (reference · medium)	[R]	full file names supplied; closing inventory cross-referenced to the closeout recompute's file

Item (review · severity)	Disposition	Where / reason
F.5 "verbatim" misnomer (reference & distillation · medium)	[R]	retitled "condensed transcription" (frame labels and marks preserved; on-disk README is the verbatim authority by fingerprint); F.1(iv) synchronized
body's map-caption quote lacks ellipsis (reference · medium)	[R]	§5 caption now carries the ellipsis with the condensation note, matching F.5
teacher-probe double naming (reference · medium)	[R]	§1.2 unified to "teacher-metadata probe"
Rule-5 third slot mislabeled (game-theory · medium)	[R]	third slot = mixed (was "uncomputability")
immunity root-cause components mis-listed (game-theory · medium)	[R]	round 1 now carries the confound component + identical-cell + resolution-floor + coverage + minus-infinity components; round 2 carries minus-infinity retention as the common primary cause (confound resolved 36/36) + resolution-floor + coverage — [**v5 erratum: this row's round-2 label miscoded the sealed verdict's own attribution — the sealed primary cause is the sampling-implementation component, with minus-infinity as the common mechanism premise; §4.2 is corrected this round (the re-verification's M3); this row's original wording is retained as the historical record of the miscoding**]
degeneration re-run line count (game-theory · medium)	[R]	one FAIL + seven PASS = eight lines
closure artifact of the fourth commit not merged (game-theory & distillation · medium)	[R]	E70 added (closure register `261ebba5c09e` / 4,789 B, commit `c247c15`); divergence C-13 updated to the post-closure caliber; conventions' push count updated to four artifacts / 4 commits / 6 files; B.4-16 HEAD chain extended
round-3 dual-metric wording (distillation · medium)	[R]	two co-reported measures: alignment-based 0.4534 against the 0.65 line, normalized-edit-distance 0.2754 against the 0.55 line; alternative readings 0.5127 / 0.2963 / 0.3636 supplied
conventions' divergence count (reference · low; distillation · low)	[R]	three divergence faces in total, with the 65-anchor pass's count (2) and the in-line dual value (1) stated separately
five-legs mapping row + independence naming (game-theory · low)	[R]	F.2 gains the independent-play-leg → GT-1-leg row with the no-collision note vs published GT-8b/8c
knife-edge cell wording (game-theory · low)	[R]	literal-and-mathematical band placement + removal/retention invariance added

Item (review · severity)	Disposition	Where / reason
M1/R1 discretion overturn clause (game-theory · low)	[R]	"registered in full so that any verifier can overturn it independently" added
path-form gaps (main-wave/run chain/README registers; game-theory · low)	[R]	prefixes and cross-references completed across E37–E62
E20 row pathless (reference · low)	[R]	`deposon_team/plugins/_v3n_cpath_live_rerun_2026_10_01.py` dual-value row
runner drift attribution (distillation · low)	[R]	authorized modification (sign-normalization), five on-disk files at today's HEAD (four at the 10-01 epoch)
closing-note positioning (reference · low)	[R]	"the closing note of the README's published-findings block"
module-table bare names (reference · low)	[R]	uniform `_september_workspace/` prefixes; re-measured all-equal this round
F.2 missing case rows (reference · low)	[R]	seven rows added (independent-play leg, near-gradient re-run, suspended-items registry, pre-registered re-run request, degeneration re-open chain, request-chain artifacts, conclusion-verification probe)
supplementary reading distribution not shown (distillation · enhancement)	[R]	E67 row carries min 0 / max 8 / population-std ≈ 2.6300 / sample-std ≈ 2.6357 + the three-artifact chain
pending-PI handling caliber (reference · low; game-theory · note)	[R]	Class D preamble now carries the PI's 2026-10-12 directive; no item is re-submitted; D-4/D-5 (submission-workflow faces) stay open
the "— II." continuation mark in the title (delegation order)	[D]	kept as approved: the PI's approval named the candidate-A literal; the mark denotes program sequence, not a main-line label (recorded in the draft-status block)
field-name caliber (`is_correction`) (distillation · low)	[D]	kept as sealed-verdict published literal; 0 rewriting of published field names; caliber note in the compliance register
the recount's compressed alternative readings (game-theory · note)	[N]	not restored to the body: the four-beat prose density follows the body-argues/appendix-lists discipline; the full alternative readings remain reachable in the verdict files and Appendix C; restoring them would reintroduce the enumeration pathology
the family's suspended-items expansion in the body (game-theory · note)	[N]	not expanded: the suspended faces are registered in Appendix D and F.2; body expansion adds no verdict content
F.2 wording drift vs the outline's mapping table (reference · low)	[N]	five phrasing variances retained: F.2's row semantics were verified distortion-free by the distillation-face review; wording is subordinated to in-file table readability

Item (review · severity)	Disposition	Where / reason
gate-prerequisite distribution attribution (C-7) (delegation order)	[W]	marked "pending team resolution, deferred this round" — the experiment side's ruling determines the entry
measurement-domain priority (shared-gate / C-2) (delegation order)	[W]	marked "pending team resolution, deferred this round" at both the body's gate paragraph and C-2

G.3 Resolution-merge round (v3, 2026-10-12)

Basis. The team-resolution register (the resolution document's §0/§1, entry numbers G-D01… G-J02, G-E09; authorized by the PI's delegation directive of 2026-10-12). The merge's sole basis is the resolution register's own literal wording; every merged item cites its entry number. The two v2 pending-resolution items ([W] in G.2) are now ruled in.

Merged item (v2 location)	Resolution entry	Disposition in v3
gate-prerequisite distribution attribution (C-7; was [W])	G-D01	ruled: authoritative distribution = the `a21915ec55a7` face (closed 13/13 — pass 7 / kd 6 / af · fail · give-up 0 each, hung 0); the `f4b129e3312f` §5 mutually-exclusive distribution retained as coexisting historical record (0 revision, 0 invalidation)
measurement-domain priority (shared-gate body paragraph + C-2; was [W])	G-D02	ruled: the full-event domain (7-day caliber) is authoritative; the legacy 3-day domain stays as co-reported contrast (its FAIL is a contrast reading, 0 re-ruling); both verdicts' final states stand
immunity round-2 effectivity (§4.2 + C-3; was B-4 open)	G-J01	ruled: effectivity established (two independent on-disk record points); the register's internal contradiction stays on file unrevised; the conditional blank is not triggered — round-2 FAIL stands
sign-convention item of the live face (§4.3; was B-7 open)	G-D09	ruled: authoritative caliber = the post-normalization runner face; historical reading keeps its registered literal; the runner-fingerprint dual value stands; five open items remain
vocabulary-expansion line (§4.6 + B-2)	G-D05	ruled: the fifteen-word table locked into the locked set; the two tier-ambiguous words carried as dual readings (both tiers read, 0 selection); the line's adjudication unlocked for the segment's future double-reading

Merged item (v2 location)	Resolution entry	Disposition in v3
missing-field backfill rule (§4.6 + B-3)	G-D04 (= G-R06)	ruled in as the held-out segment's execution caliber; the collection verdict's KD final state unchanged (the rule applies to the segment's new readings, not to on-file readings)
matrix-attribution item (C-6; bang3 chain R-01)	G-D10	ruled: scope-of-application interpretation, append-only registration — the nominal three-valued axis falls outside the numeric-degradation alert's per-field scope (n_distinct = 3 is full-class coverage, not collapse); the three affected family verdicts unchanged, 0 criterion-text amendment

Merged item (v2 location)	Resolution entry	Disposition in v3
distortion-bound request items + theory-source delivery (§4.3 + A-1/A-2 + the P-B slot)	G-D03	ruled: request re-confirmed, channel re-routed to an external derivation commission under the theory-source caliber (item-by-item verifiable delivery, team acceptance before landing), normalization constant per the spec's registered literal, landing assigned post-acceptance, adversarial face queued on genuine inputs; the slot remains pending-external-input; the team creates 0 theoretical content and pre-fills 0 numeric bounds — appended (v8, the PI's adoption ruling of 2026-10-12 21:54): the external derivation delivered (the reply in the repository's letters/, in force at remote HEAD `74c7d76`; the E74 row), and its forward payoff values are adopted value-and-condition together (u* = 1 in 15 cells, max_u = 1 in the five families; the numerical minima and tolerances remain null); the slot advances pending-external-input → adopted; the empirical instantiation remains un-run; the re-routed commission line is superseded by the adoption ruling, while the landing-side conditions (team acceptance before landing, the calculator-and-anchors on the direct-execution line, the adversarial face on genuine inputs) stand — appended (v9): the acceptance-side condition of that landing set is discharged — the delivered mathematics' acceptance verification (independent re-derivation and finite enumeration, registered locally pending repo anchoring) passed on 2026-10-12 — so the landing's live gate is the pre-registration freeze (the calculator-and-anchors land as the freeze's components on the direct-execution line); the adversarial face stays queued on genuine inputs

Merged item (v2 location)	Resolution entry	Disposition in v3
first-paper introduction supply caliber (§2.5)	G-E09	ruled: no new supplementary-experiment round; inherited readings supplied by on-disk recomputation under the three-artifact citation discipline (0 pre-run)
recount's four hung items (B-5)	G-J02	held on file: the authorization fact-layer is unverifiable on disk; quoted with an unverifiable mark; 0 fabricated authorization
gate closure-not-release caliber (§4.6 + B-1)	G-D08	confirmed: closure-is-not-release stands; the three release prerequisites are in place per the resolution set (E68 consolidation, G-D05 lock, G-D08 confirmation); the release action remains a literal-source-checked act — 0 release claimed in this paper

Ruling-literal discipline. Every merged ruling is transcribed from the resolution register's own wording (0 paraphrase of the ruling content, 0 new values beyond the register's registered ones); no verdict literal is altered (all merged items are annotation-layer or status-layer — the register's own red lines: 0 reversal of existing verdicts, 0 threshold retro-fit, 0 fabricated evidence). De-labeling is preserved: the resolution entry numbers (G-D…) appear as citation anchors only, matching the registered-artifact citation convention.

G.4 Local-reading merge round (v4, 2026-10-12)

Basis. The executing seat's local execution wave (registration §8, append-only): one supplementary adjudication fully chain-closed (pre-registration + verdict), local artifacts pending push (pushed since at commit `74c7d76` — G.7); one pre-registration-only freeze on a second item, which per the commissioning instruction is **not merged into this draft** (0 mention beyond this line). New-anchor fingerprints are the registration's values (a80d9ae8008b / 5,678 B; 338f30ad7460 / 6,156 B), re-measured in place by this seat (marked [m] at the value tails of the E71 row) (the pair sits untracked in the workspace, pending the next upload window — v9 note: superseded, pushed at `74c7d76` per the G.7 note above; the round-time literal retained).

Merged item (v3 location)	Source	Disposition in v4
vocabulary-expansion supplementary adjudication (§4.6 sentence tail; the G-D05-unlocked line)	local chain, registration §8	ruling transcribed verbatim-semantics from the verdict file: hit line fires (`k_v3s_3_4_hit = True`, FAIL at the criterion's hit line); four dual-caliber readings (base thirty-six-word face 0.571429 / 0.500000; locked thirty-nine-word face 0.714286 / 0.700000, all $<$ 1.00); twelve registered readings, none $\geq 1.00 \Rightarrow$ conclusion fully robust; arithmetic self-check 12/12; main verdict SHA-rechecked identical before/after (0 touching, 0 reversal, 0 retro-edit); three dual-value faces co-reported 0 resolution; root cause = construction-face beta component, partially repaired (indexes 47/50 genuinely missing + structural ceiling, 1.00 unreachability as on-file literal)
abstract + contribution A6 + audit-map substrate row	consistency sweep	fourth-line firing synchronized (twelve readings below coverage; local chain; main verdict untouched)
Appendix A	new anchor	E71 row: the local pair with pending-push caliber (0 fabricated remote path); conventions and B.4-15 updated to the five-anchor-row count — the pair was pushed at `74c7d76` after this round (G.7)
Appendix D B-2	status sync	ruled by G-D05 and closed by today's supplementary adjudication (local chain, E71, pending push at this round — pushed since at `74c7d76`, G.7)
the pre-registration-only freeze item	commissioning instruction	not merged: 0 content citation, 0 status claim beyond this G.4 line (execution sits with the next window; the paper neither reports nor schedules it)

Ruling-literal discipline (this round). The hit-line expression (`k_v3s_3_4_hit = True`) and the four readings are transcribed from the verdict file's registration (the identifier appears as a registered-line citation, matching the verdict-line map in F.3); the main verdict's KD final state stands (0 touching, 0 reversal); dual readings co-reported (three dual-value faces); 0 new un-measured values (the E71 pair re-measured in place; all other values are prior rounds' registered

ones).

G.5 Re-verification repair round (v5, 2026-10-12)

Basis. The re-verification pair — the mechanical-gate closing report (pass: 71 anchors all-true, all 30 blocking items repaired; 3 low + 2 observations, non-blocking) and the resolution-verification review (content layer: zero drift; paper-surface: 5 medium items, disposed item-by-item below against the review's position/problem/change-to triples).

Item (verification basis)	Disposition in v5
M1 — cross-day-domain row not upgraded with the ruling (contradiction with C-2/§4.6's ruled state)	C-1 row's status column upgraded to the G-D02 ruling: whole-event domain (7 days ⇒ PASS) authoritative; old accounting domain (3 days ⇒ FAIL) co-reported contrast, 0 re-ruling; the no-softened-phrasing discipline clause retained
M2 — live-sign-caliber row not upgraded	C-24 row's status column upgraded to the G-D09 ruling: post-normalization runner face authoritative; historical literal kept; E20 dual value maintained
M3 — immunity round-2 primary cause mislabeled (inherited from v2's repair table)	§4.2 corrected to the sealed verdict's own attribution: the sampling-implementation component is the common primary cause (same item, different draws, different decoys — α spread = implementation drift, not immunity breach); minus-infinity retained as the common mechanism premise (confound resolved 36/36); G.2's row carries an inline v5-erratum note (original wording retained as the historical record of the miscoding)
M4 — mapping row conflating the published near-gradient face with the low-arm re-runs	F.2 row renamed "the low-arm re-run leg" mapping to the lowarm r1/r2 verdicts (gt8), with the near-gradient leg (published GT-6) explicitly assigned to §2.5
M5 — the two gate-distribution faces absent from the evidence ledger	E72/E73 rows added: true paths + dual values (git-traceable historical blob `a21915ec55a7`/31,672 B and registered `f4b129e3312f`/14,770 B vs current-disk `3b00b6377dab`/31,652 B and `22e48303c8ad`/14,750 B, both re-measured in place this round, matching the re-verification's measured values); value-level readings identical across faces and unaffected by the correction-wave drift; C-7 row cross-references E72/E73; conventions updated
L1 — leg-name inconsistency (independence vs independent-play)	body §4.1 (two places) + audit-map row unified to "the independent-play leg" (the re-verification's recommended precise form)
L2 — the five-files count's unverifiable sub-clause	C-26 and §4.3 restated under the string-search rule (five carriers at today's HEAD, the counting artifact excluded) with the register's chain-face caliber co-stated (five faces, two carrying the post-modification value)
L3 — capitalization slip in the line-map	corrected (the vocabulary-expansion hit-flag entry's follow-on word lowercased)

Item (verification basis)	Disposition in v5
mechanical-gate low items — divergence-count ambiguity / stray bracket / same letter-form	divergence counts already disambiguated in the conventions (two denominators: the 65-anchor pass's 2 vs this ledger's 3 divergence faces); the stray-bracket residue was repaired in the v2 round for the v2-era rows; two marks introduced by the v4 round (the E71 row's and the G.4 paragraph's) remained free-floating and are positioned to the value-tail form this round (v6) — this row's earlier "re-scanned 0" claim is corrected to that actual state; the S4/S5 same letter-form note is now explicit in F.4
observations (2, registered non-blocking)	T1 adopted: the G-E06 sentence gains the explicit "no re-run of the executor — registered readings adopted read-only" clause; T2 registered: the resolution-side row-referencing phrasing is aligned to "Appendix C row C-1" (the resolution-document errata themselves belong to the resolution seat's next round); the E71-pending-push re-run of the closing gate stays scheduled after the push window (the push has since landed at `74c7d76`; the closing-gate re-run moves to the next full re-measurement round — G.7)
verification-side base issues (2, not paper defects)	registered for hand-over: the resolution document's value-face attribution notes and its execution-landing phrasing (the re-verification's B1/B2/T2 recommendations) — the paper carries face designations and counts only, 0 inherited mis-attribution

Discipline re-run. Mechanical gate: body 0 substantive hits (the sole hit is the whitelisted corpus graph letter-forms inside published GT-6 readings — F.4's note now explicit). Verdict literals 0 softening (M1/M2 are status-column upgrades to already-ruled states; M3 is a correction **to** the sealed verdict's own attribution, un-softening it). Dual readings co-reported (E72/E73 dual values; C-26 dual calibers). 0 new un-measured values (E72/E73 re-measured in place, matching the re-verification's values).

G.6 Low-item closing round (v6, 2026-10-12)

Four items disposed (minimal diff, direct关联句 only): the E72/E73 rows' "value-level" corrected to **count-level** with the C-7 swapped-pair note (the re-verification's low 1); the E72 row's "git blob `df51dc3`" corrected to "the historical blob at commit `df51dc3` (the 10-12 upload batch carrying the 10-11 artifact)" (the gate report's low 1); the E71 row's and G.4's free-floating [m] marks positioned to the value-tail form and the G.5 "re-scanned 0" claim corrected to the actual state (the gate report's low 2); the conventions' divergence sentence extended to the seven dual-value rows with the two-denominator caliber stated and the five-anchor sentence + B.4-15 itemized (the gate report's low 3). Optional enhancement adopted with reason: the C-7 row gains the resolution-erratum cross-reference half-sentence (the re-verification's note 1) — the ruling's face designation is unchanged by the erratum, and the half-sentence closes the citation chain at the paper's own row.

G.7 Post-push caliber upgrade round (v7, 2026-10-12)

One-line disposition: the push window landed (remote main `c247c15` → `5e865e3` (a four-file letters batch) → `74c7d76` (the E71 pair and the held-out segment's pre-registration, add-only), verified by `git ls-remote`; the E71 pair and the segment pre-registration re-measured in place at the remote HEAD this round, all equal), so the E71 anchor row and every live pending-push caliber (the conventions, B.4-15, B-2, the §4.6 sentence, the held-out segment's status sentence, A-5) are upgraded to the pushed caliber, and the round-history rows (G.4 ×3, G.5 ×1) carry appended time-point notes rather than rewrites (append-only). 0 new un-measured values; 0 verdict-literal changes; F.3 checked for pending-push phrasing — 0 occurrences (the row cites the verdict file, not its push state).

Self-checks re-run for this round (v7). De-labeling mechanical gate: body (§1–§7) 0 hits on the identifier family and the stage-label family (the header's logistics sentence stays inside the to-be-stripped status block). Verdict literals untouched (0 softening; all repairs are path-layer and precision-layer). Dual readings co-reported throughout (E20/E26/E28 dual-value rows retained). 0 new un-measured values introduced: every [m] mark is this seat's own measurement at HEAD `c247c157eced...{}` (v9 dual-HEAD note per the closing report's low 1: that literal covers the v2-round marks; the E71 pair's value-tail [m] marks and the E74 row's [m] mark are re-measurements in place at the remote HEAD `74c7d76` — the v7 and v8 rounds respectively; E74 re-verified this v9 round, values unchanged).

G.8 Adopted forward-value merge round (v8, 2026-10-12)

Basis. The PI's adoption ruling of 2026-10-12 21:54 (the codex unblock reply's forward values adopted: u* = 1 in 15 cells, max_u = 1 in the five families; the numerical minima and tolerances remain null; a value-and-condition adoption) + the review's §1.4 mapping (the reply's affected faces: §4.3, A-1/A-2, the G-D03 disposal row; 0 new conflicts; the three corrected claims of the 09-30 reply 0 carried by the then-current draft) + the reply itself (the repository's letters/, in force at remote HEAD `74c7d76`; the same-batch codex trio clone-verified in place).

Location (as in v7)	Disposition in v8
§4.3 distortion-bound paragraph	the adopted values merged with the applicability conditions carried inseparably (binary scoring of accepted-versus-rejected decisions; per-decision full-information oracle comparator with the restricted-information optimum not asserted equal and sequential-game regret read at the actual state; the non-empty evaluator-key premise; the game-decision family's criterion-key specification (best-response vs equilibrium recognition, cardinal-loss discard); the complete-evaluation family's full-information caliber); T-2/T-3 null + blockers carried as-is (0 claims an established bound); the two operational definitions (information-channel capacity in bits per channel use; payoff-relevant identifiability, the conditional common-action criterion), the conditional T-2 lower-bound derivation (the exact minimum for the stated exogenous class), and the uniform-mechanism exact check (conditional error $1 - |K_t|/4$; $3H/4$ under singleton keys) registered as **adopted external derivation content** under the theory-source caliber (the C4 correction register: the theory-bound source is a codex derivation); the slot pending-external-input → adopted; the empirical instantiation remains un-run
App D A-1/A-2	status advanced to adopted (PI 2026-10-12 21:54); the empirical instantiation and the post-acceptance landing stay un-run (v9 micro-sync: the acceptance verification passed, the landing gate advanced to the pre-registration stage); the G-D03 five-item wording upgraded in place
G.3's G-D03 row	appended time-point note (append-only): the external delivery arrived; the forward values adopted value-and-condition together; the slot advances; the re-routed commission line superseded by the adoption ruling, the landing-side conditions stand
abstract + contribution A3 + audit-map live row + §4.3 bridge + §7 conclusion sentence	consistency sweep on the blocked/undelivered family: "blocked on a missing theory input" → "blocked, its forward payoff values adopted as design inputs and its theoretical bounds still undelivered" (and equivalent re-basings); the blocked verdict literal itself unchanged; the sweep's contribution-level target is A3 (A6's substrate text carries 0 related wording, checked)

Appendix A + conventions + B.4-15 + F.2	one new anchor row (E74: the letters/ reply, re-measured in place at the remote HEAD this round, the same-batch trio co-verified); the letters/ path convention, the added-today count, and B.4-15 extended; one F.2 row added for the adopted forward values

Self-checks re-run for this round (v8). De-labeling mechanical gate (quick scan): body (§1–§7 + abstract) 0 substantive hits — `u*`/`max_u` are the spec's mathematical symbols (`u*(a_t)` already present in v7's body; `max_u` named in the spec and referred to descriptively in v7's body, its symbol string entering the draft with the adopted values — precision-fixed v9 per the closing report's low 2), the ruling date and the remote-HEAD hash follow the existing citation convention (commit hashes already appear in the body), and 0 slot labels, 0 commission identifiers, 0 letters/ paths, 0 not-yet-registered acceptance-line references appear in the body. Verdict literals 0 softening: the blocked verdict stands verbatim ("blocked, in full"); T-2/T-3 carried as null; the "not arrived" clause is re-based to the undelivered bounds — a status-layer update to the PI-ruled state, not a verdict change. 0 new un-measured values: the adopted u*/max_u values are transcribed from the adopted external derivation file (external-derivation-content transference), and the artifact's fingerprint is re-measured in place at the remote HEAD this round (the [m] mark at the E74 row's value tail, the E71-row form — this seat's in-place re-verification; the same-batch trio co-verified; v9 mark-form note: the v8 file's plain `]` at that tail is normalized to the [m] form this round so the byte-level form matches the E71 row and the conventions' [m] definition). No-upgrade boundary held: the adopted values appear nowhere as calibrated findings or established bounds (the forward-design-input framing is stated at every landing site; "an adopted forward value is not a delivered bound"). The three codex-corrected claims of the 09-30 reply (the aggregate range [0, H] not [0, 1]; the guarded, non-unconditional normalizer cancellation; the identifiability threshold) — full-text scan of this draft: 0 carried (re-verified this round, matching the review's registration; any future citation of that reply's mathematical content follows the corrected caliber). 0 claims the theoretical bound is established (the minima and tolerances remain null as null).

G.9 Fidelity-precision and micro-sync round (v9, 2026-10-12)

Basis. The theory-face fidelity check (`4473311e9d4c` — medium 1 / low 6 / hints 5, the medium copy pre-staged for direct use) + the incremental closing report (`e4a2601e106d` — gate passed; 4 remaining low items, non-blocking) + the acceptance-line registration (`2e7d804dad2a` — the delivered mathematics' acceptance verification: four objects all PASS by independent re-derivation and finite enumeration, 0 trust in external scripts; a process-layer registration, local pending repo anchoring).

Location (as in v8)	Disposition in v9
§4.3 binary-scoring tail (the fidelity check's medium 1)	the reply's reject-not-relabel boundary merged with the pre-staged copy: "— adoption would choose that loss meaning rather than fill a harmless numerical scale slot; if the intended measure must retain cardinal game losses, this binary proposal is rejected for those cells rather than relabeled as that measure" (the registered boundary's paper-side counterpart: the binary-element case is rejected for those cells, not relabeled)

Location (as in v8)	Disposition in v9
§4.3 five low-level precision restorations	the non-empty-key clause's mechanism cases restored to the letter's three (removes actions / changes their outcomes / lets the key depend on its internal state; that state and feasibility must be included and nonemptiness checked before the values apply); the second negation extended to "about any model or mechanism"; the conditional derivation's class wording restored to the admissible observation-based policy class (deterministic choices attain it and randomization cannot improve); the uniform-check boundary hedged to "not automatically the information-constrained minimum"; the identifiability boundary restored to "need not identify the whole mechanism, state, or a unique label"
§4.3 theory-source ledger parenthesis	the re-review status extended with the current time-point: the 2026-10-12 fidelity check verified this draft's source-caliber statements against the correction register with 0 divergence, and the adoption ruling of the same day proceeded on this caliber (the register's own pending-re-review literal retained as history)
the four post-acceptance sites (A-1, A-2, the G-D03 row, the G.8 map row)	micro-synced per **option (a)**: the delivered mathematics' acceptance verification passed on 2026-10-12 (independent re-derivation and finite enumeration, registered locally pending repo anchoring), so the landing's live gate advances from post-acceptance to the pre-registration freeze (the calculator and five-anchor pre-registration land as the freeze's components on the direct-execution line; the empirical instantiation stays un-run; the adversarial face stays queued on genuine inputs); the G-D03 row carries an appended v9 time-point note (append-only), the other three sites carry in-place v9 markers. **Reason registered**: the packaging round follows immediately — keeping the stale pending-acceptance reading (option b's wait-for-anchoring) would ship it into the arXiv package, while the acceptance state is a process-layer fact statable with its local-registration provenance (the same provenance form the draft already uses for the adoption ruling itself); the body keeps 0 acceptance-line references

Location (as in v8)	Disposition in v9
G.7 self-check / G.8 self-check / G.4 tail parenthesis (the closing report's low 1–4)	the G.7 sentence's HEAD literal scoped with a dual-HEAD note (the v2-round [m] marks at `c247c157eced...{}`; the E71 pair's and E74's [m] marks re-measured in place at the remote HEAD `74c7d76`; E74 re-verified this round, values unchanged); the G.8 self-check's symbol-layer half-sentence precision-fixed (`u*(a_t)` present in v7's body; `max_u` named in the spec, referred to descriptively in v7's body, its symbol string entering with the adopted values); the §4.3 re-review note (the row above); the G.4 tail parenthesis gains a supersession time-point note (pushed at `74c7d76` per G.7; the round-time literal retained) — text-layer items all disposed this round; 0 pure-typography items found left for the typesetting round
Appendix A E74 row (mark-form normalization, found in this round's byte-level audit)	the v8 file's plain `]` at the E74 row's value tail normalized to the [m] form (the E71-row byte-level form; the conventions' [m] definition) — the registered round records describe the mark in display-equivalent form, and this normalization makes the byte-level form match them; the G.8 self-check's mark phrase corrected accordingly

Hints (5), each registered. (1) "no existing Nash label supplies either key by itself" — not adopted: the criterion-key clause already carries the meaning (the fidelity check itself reads it as covered). (2) The same-scoring-same-key consistency sentence — not adopted: outside the adoption's minimal merge face; the content lives in the acceptance-line registration's basis. (3) The no-numerical-selection / no-performance-measurement sentence — not adopted: the adjacent clauses carry the no-measurement discipline (the IC-capacity note's "does not assign a measured channel"; the forward-design-input triple negation). (4) The random-horizon extension and the null-L four-case enumeration — not adopted: the former exceeds the adopted minimal face; the latter is carried in the acceptance registration's re-derivation side. (5) The caution sentence's translation note — **adopted at the register layer, here**: the register's Chinese literal 「来源已改」≠「理论已到位」 renders as the quoted *a changed source is not a delivered theory* (the §4.3 sentence carries the English rendering; the register file remains the verbatim authority, anchored in Appendix A).

Self-checks re-run for this round (v9). De-labeling mechanical gate (quick scan): body (§1–§7 + abstract) 0 substantive hits (the merged clauses introduce 0 identifiers, 0 labels, 0 paths; the four micro-sync sites are appendix-layer; the fidelity-check and closing-report fingerprints in the G.9 basis sit in the round-history layer). Verdict literals 0 softening: blocked (6 sites) unchanged; null as null; the medium-1 clause and the five restorations are boundary-tightenings (reject-not-relabel, hedged minimum, need-not-identify), not softenings. 0 new un-measured values: every merged phrase is a transcription from the verified source letter, the pre-staged copy, or a registered basis file; the E74 anchor re-verified this round with values unchanged (the mark-form normalization changes 0 values). No-upgrade boundary held: the added clauses tighten the adoption's conditions; 0 calibrated-findings or established-bound claims introduced. The 09-30 reply's three corrected claims: 0 carried (re-scanned). Diff surface: 10 regions (listed in the delivery note), 0 other changes.

End of draft v9 (fidelity-precision and micro-sync round; dispositions by round: G.2 repair, G.3 resolution merge, G.4 local-reading merge, G.5 re-verification repair, G.6 low-item closing, G.7 post-push caliber upgrade, G.8 adopted forward-value merge, G.9 fidelity-precision and micro-sync). Prepared 2026-10-12 by the paper's drafting seat (identity withheld per the anonymous-draft convention; the single-arc structure follows the approved outline v3; this v9 is a new file — v8 through v1 and the archived dual-track drafts remain on disk untouched; the item-by-item dispositions are Appendix G.2–G.9). All content follows the project's external caliber; this file's own fingerprint is measured after writing and reported in the delivery note, not written into the file.

AI use disclosure

The framework and the research direction were provided by the principal investigator, the criteria were pre-registered before the runs, and every disposition was adjudicated under the principal investigator's direction. This manuscript was then prepared with AI assistance, and the AI participation is stated in full. The drafting model is GLM; the manuscript was drafted, reviewed, and verified by a multi-agent pipeline under that direction. All experimental readings, verdict literals, thresholds, counts, and fingerprints reported in this paper are transcribed from sealed on-repository artifacts and were mechanically re-verified; they are not AI-generated content. AI participation went beyond drafting: the V4 (distillation) line of this work was created in a multi-agent brainstorm in which agents of every role in the team took part, together with Workbuddy, Doubao, Qianwen, and coze, among others; each agent spawned its own sub-agents to brainstorm independently, and minimax consolidated the results. The experimental line was run in the minimax code client: the main agent dispatched sub-agents to execute the experiments, the sub-agents being all minimax models, while the main agent itself used minimax, MiMo, and DeepSeek, among others. Codex and Claude both participated in the theoretical derivations, and the simple reasoning was completed by minimax. Bug fixing during development was handled by GLM through the Trae client. The PDF files used for partner contact and reporting were handled by coze, and upload-side operations were handled with the assistance of Kimi. This work builds on the program's first paper, whose foundation (the core algorithm, the experimental pipeline, the data-analysis scripts, and the first draft) was produced with the assistance of KIMI-K3 under the author's direct instruction, as that paper's own disclosure records. Across these faces, work was transmitted and coordinated between agents, with letters as an important form of that transmission and coordination. The draft was examined by multiple independent rounds of machine review, with each finding adjudicated by the team and its disposition recorded in the appendices' governance and repair-round registers (Appendices B and G) rather than silently absorbed; each number in the text maps to its sealed artifact, cited in place and registered in the evidence ledger (Appendix A); the program's public repository is `github.com/zeroandcat/Deposon`. The author reviewed the manuscript and takes full responsibility for the entire content of this paper.

References

[R0] The first paper of the Deposon program (arXiv:2609.09001, cs.AI; the repository's published mind-map and game-theory-explanation phases; on-disk authority copy: `paper/deposon_paper_final_en.md` = `2ec20b96e941` / 68,164 B). Described as an arXiv preprint; 0 peer-review record, and 0 "published/accepted/merged" wording is used anywhere in this draft for it.

[R1] Monderer & Shapley, “Potential Games,” Games and Economic Behavior, 1996.

[R2] Rosenthal, “A Class of Games Possessing Pure-Strategy Nash Equilibria,” IJGT, 1973.

[R3] Sandholm, “Evolutionary Game Theory and Multi-Agent Learning,” 2010.

[R4] Koutsoupias & Papadimitriou, “Worst-case Equilibria,” STACS 1999.

[R5] Roughgarden & Tardos, “How Bad is Selfish Routing?,” JACM 2002.

[R6] Christodoulou & Koutsoupias, “The Price of Anarchy of Finite Congestion Games,” STACS 2005.

[R7] Roughgarden, “The Price of Anarchy is Independent of the Network Topology,” STOC 2002.

[R8] Roughgarden, Selfish Routing and the Price of Anarchy, MIT Press, 2005.

[R9] Nisan et al. (eds.), Algorithmic Game Theory, Cambridge University Press, 2007.

[R10] Nisan & Ronen, “Algorithmic Mechanism Design,” STOC 1999 / Games Econ. Behav. 2001.

[R11] Conitzer & Sandholm, “Complexity of Mechanism Design,” UAI 2002.

[R12] Procaccia & Tennenholtz, “Approximate Mechanism Design Without Money,” EC 2009.

[R13] Fan, C., et al., “Can Large Language Models Serve as Rational Players in Game-Theoretic Tasks?,” AAAI 2024 (arXiv:2312.05488).

[R14] Mao, S., et al., “ALYMPICS: LLM Agents Meet Game Theory,” ACL 2025.

[R15] Lorè, N., et al., “Strategic behavior of large language models and the role of incentives,” Scientific Reports, 2024, article 18490 (arXiv:2309.05898).

[R16] Zarian et al., agent-interaction frameworks (as registered in the scan; exact titles pending verification — not cited in the body text beyond the scan’s face).

[R17] (Reserved slot — the scan’s registered placeholder; no entry assigned.)

[R18] Weng et al., “Verifiable Machine Learning” (survey; as registered).

[R19] Sun et al. (Proof-of-Learning) (as registered in the ledger’s scan; title per ledger literal).

[R20] Wei et al., “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models,” NeurIPS 2022.

[R21] Wang et al., “Self-Consistency Improves Chain of Thought Reasoning in Language Models,” ICLR 2023.

[R22] Zhou et al., “Least-to-Most Prompting,” ACL 2023.

[R23] Kojima et al., “Large Language Models are Zero-Shot Reasoners,” NeurIPS 2022.

[R24] Mirzadeh et al., “GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models,” ICLR 2025 (eprint 2410.05229).

[R25] Nezhurina et al., “Alice in Wonderland: Simple Tasks Showing Complete Reasoning Breakdown in State-of-the-Art Large Language Models,” arXiv 2406.02061, 2024.

[R26] Lanham et al., “Measuring Faithfulness in Chain-of-Thought Reasoning,” TMLR 2023.

[R27] Turpin et al., “Language Models Don’t Always Say What They Think,” NeurIPS 2023.

[R28] Berglund et al., ICLR 2024 (Reversal Curse).

[R29] McNemar 1947 (statistical note on the comparison of paired proportions).

[R30] Lipton & Steinhardt, CACM 2019.

[R31] Dodge et al., EMNLP 2019.

[R32] Recht et al., ICML 2019.

[R33] Tevet & Berant, EACL 2021.

[R34] D’Amour et al., JMLR 2022.

[R35] Lazaridou et al., NeurIPS 2021.

[R36] (Reporting-reform entry as registered in the scan; details pending verification — cited only at the face level.)

[R37] (Registered scan entry; pending verification — not cited beyond the genre face.)

[R38] Saxena et al., “Improving Multi-hop Question Answering over Knowledge Graphs using Graph Summarization,” ISWC 2023.

[R39] Wang et al., “Benchmarking Commonsense Knowledge Graph Completion,” 2023.

[R40] Cui et al., “Zeshel: Bridging Entity Types across Datasets,” 2019.

[R41] Dziri et al., “Evaluating the Ground Truth of Multi-hop Reasoning” (as registered).

[R42] Hinton et al., “Distilling the Knowledge in a Neural Network,” 2015.

[R43] Gou et al., “Knowledge Distillation: A Survey,” IJCV 2021.

[R44] Ho et al., “Large Language Models Are Reasoning Teachers,” ACL 2023.

[R45] Magister et al., “Teaching Small Language Models to Reason,” ACL 2023.

[R46] Wang et al., “Self-Instruct: Aligning Language Models with Self-Generated Instructions,” ACL 2023.

[R47] Tian et al., “Scalable Soft-Label Supervised Contrastive Learning,” 2023 (on-policy distillation face).

[R48] Sharma, M., et al., “Towards Understanding Sycophancy in Language Models,” ICLR 2024 (arXiv:2310.13548).

[R49] Batzner, J., “Sycophancy Claims about Language Models: The Missing Human-in-the-Loop,” ICLR 2025 Workshop on Bidirectional Human-AI Alignment (tiny paper) (arXiv:2512.00656).

[R50] Fan, C., et al., AAAI 2024 (cross-listed with the game-theory band; rational-player face).

[R51] Mao, S., et al., ACL 2025 (cross-listed; agents-and-game-theory face).

[R52] Lorè, N., et al., Scientific Reports 2024 (cross-listed; strategic-behavior face).

[R53] Hofman, J. M., et al., "Pre-registration for Predictive Modeling," arXiv:2311.18807, 2023.